Problem 1 (Broken IND-CPA to IND-CCA transformation).

Definition. A MAC system is a pair of PPT algorithms Sign, a signing algorithm, and Vrfy, a verification algorithm.

(a) Sign takes as input a key k and a message m and output a tag t.

(b) Vrfy is a deterministic algorithm that takes as input a key k, a message m and a tag t and outputs accept or reject.

We require that tags generated by Sign are always accepted by Vrfy; that is, the MAC must satisfy the following correctness property: for all keys k and for all messages m,

$$\Pr[\mathsf{Vrfy}_k(m, \mathsf{Sign}_k(m)) = \mathsf{accept}] = 1.$$

As usual, We denote by $\mathcal{K}$ the key space, $\mathcal{M}$ the message space and $\mathcal{T}$ the tag space.

Definition (MAC security with verification queries). For a given MAC system $\mathcal{I} = (\mathsf{Sign}, \mathsf{Vrfy})$, the MAC experiment runs as follows:

(a) The challenger picks a random $k \xleftarrow{\$} \mathcal{K}$.

(b) $\mathbf{Adv}$ queries the challenger several times. For $i = 1, 2, \ldots$, the ith signing query is a message $m_i \in \mathcal{M}$.

(c) Given m_i the challenger computes a tag $t_i \leftarrow \mathsf{Sign}_k(m_i)$ and returns t_i to $\mathbf{Adv}$.

(d) Eventually $\mathbf{Adv}$ outputs a candidate forgery pair $(m, t) \in \mathcal{M} \times \mathcal{T}$ that is not among the signed pairs, i.e.,
$$(m, t) \notin \{(m_1, t_1), (m_2, t_2), \ldots\}.$$

(e) $\mathbf{Adv}$ wins iff the challenger ever responds to a verification query with accept, i.e., $\mathsf{Vrfy}_k(m, t) = \mathsf{accept}$.

Definition. We say that a MAC system $\mathcal{I}$ is secure if for all PPT adversary $\mathbf{Adv}$, there is a negligible function μ such that in the above experiment MAC

$$\Pr[\mathbf{Adv}\text{ wins}] \leq \mu(n).$$

Consider the following attempt at transformation a CPA-secure scheme to an IND-CCA-secure one. Let $(\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ be an IND-CPA-secure PKE defined over $\mathcal{K} \times \mathcal{M}, \mathcal{C}$ and let $(\mathsf{Sign}, \mathsf{Vrfy})$ be a secure MAC with key space $\mathcal{K}$. We construct a new PKE scheme $\mathsf{PKE}' = (\mathsf{KeyGen}', \mathsf{Enc}', \mathsf{Dec}')$ with message space $\mathcal{M}$, as follows:

$\mathsf{KeyGen}'(1^n)$

```
return KeyGen(1^n)
```

$\mathsf{Enc}'_{\mathsf{pk}}(m)$

```
k <-$ K
ct <- Enc_pk(k, m)
t <- Sign_k(ct)
return ct' = (ct, t)
```

$\mathsf{Dec}'_{\mathsf{sk}}(ct' = (ct, t))$

```
(k, m) := Dec_sk(ct)
if Vrfy_k(ct, t) = accept then
    return m
else
    return ⊥
```

One might expect this scheme to be IND-CCA-secure because a change to a ciphertext (ct, t) will invalidate the MAC tag t. Show that this is INCORRECT. i,e,. show an IND-CPA-secure PKE scheme $\Pi = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ for which $\Pi' = (\mathsf{KeyGen}', \mathsf{Enc}', \mathsf{Dec}')$ is not IND-CCA-secure (for any choice of MAC).

Answer. To show this, we provide a counterexample.
Let $\Pi^0 = (\mathsf{KeyGen}^0, \mathsf{Enc}^0, \mathsf{Dec}^0)$ be a δ-correct, γ-spread, IND-CPA-secure PKE scheme over $(\mathcal{K}\times\mathcal{M}, \mathcal{C})$, where δ and $2^{-\gamma}$ are negligible. Also, let $\mathcal{I} = (\mathsf{Sign}, \mathsf{Vrfy})$ be a secure MAC system with message space $\mathcal{C} \times \mathcal{C}$, key space $\mathcal{K}$, and tag space $\mathcal{T}$.

Fix $k_n^0 \in \mathcal{K}_n$ and $m_n^0 \in \mathcal{M}_n$ for each $n \in \mathbb{N}$. We construct an IND-CPA-secure PKE scheme $\Pi = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ over $(\mathcal{K} \times \mathcal{M}, \mathcal{C} \times \mathcal{C})$ as follows:

$\mathsf{KeyGen}(1^n)$	$\mathsf{Enc}_{\mathsf{pk}}(k, m)$	$\mathsf{Dec}_{\mathsf{sk}}(ct_1, ct_2)$

return $\mathsf{KeyGen}^0(1^n)$	**return** $(\mathsf{Enc}^0_{\mathsf{pk}}(k, m_n^0), \mathsf{Enc}^0_{\mathsf{pk}}(k_n^0, m))$	**if** $\mathsf{Dec}^0_{\mathsf{sk}}(ct_1) = \perp \vee \mathsf{Dec}^0_{\mathsf{sk}}(ct_2) = \perp$ **then**
		    return $\perp$
		$(k_1, m_1) := \mathsf{Dec}^0_{\mathsf{sk}}(ct_1)$
		$(k_2, m_2) := \mathsf{Dec}^0_{\mathsf{sk}}(ct_2)$
		if $m_1 \neq m_n^0 \vee k_2 \neq k_n^0$ **then**
		    return $\perp$
		return (k_1, m_2)

We claim that $\Pi = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ is indeed correct and IND-CPA-secure.

Claim 1.1. $\Pi = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ is a correct PKE scheme. Specifically, Π is 2δ-correct.

Proof. Since the PKE scheme $\Pi^0 = (\mathsf{KeyGen}^0, \mathsf{Enc}^0, \mathsf{Dec}^0)$ is δ-correct, for each $n \in \mathbb{N}$ and any $(k, m) \in \mathcal{K}_n \times \mathcal{M}_n$,

$$\Pr\left[\mathsf{Dec}^0_{\mathsf{sk}}(\mathsf{Enc}^0_{\mathsf{pk}}(k, m)) \neq (k, m)\right] \leq \delta, \tag{1.1}$$

where the probability is taken over the randomness of $\mathsf{Enc}^0_{\mathsf{pk}}$ and $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}^0(1^n)$.

Therefore, for each $n \in \mathbb{N}$ and any $(k, m) \in \mathcal{K}_n \times \mathcal{M}_n$, we deduce by union bound that

$$\begin{aligned}
\Pr\left[\mathsf{Dec}_{\mathsf{sk}}(\mathsf{Enc}_{\mathsf{pk}}(k, m)) \neq (k, m)\right] &= \Pr\left[\mathsf{Dec}^0_{\mathsf{sk}}(\mathsf{Enc}^0_{\mathsf{pk}}(k, m_n^0)) \neq (k, m_n^0) \vee \mathsf{Dec}^0_{\mathsf{sk}}(\mathsf{Enc}^0_{\mathsf{pk}}(k_n^0, m)) \neq (k_n^0, m)\right] \\
&\leq \underbrace{\Pr\left[\mathsf{Dec}^0_{\mathsf{sk}}(\mathsf{Enc}^0_{\mathsf{pk}}(k, m_n^0)) \neq (k, m_n^0)\right]}_{\leq \delta \ \ (\text{by } (1.1))} + \underbrace{\Pr\left[\mathsf{Dec}^0_{\mathsf{sk}}(\mathsf{Enc}^0_{\mathsf{pk}}(k_n^0, m)) \neq (k_n^0, m)\right]}_{\leq \delta \ \ (\text{by } (1.1))} \\
&\leq 2\delta,
\end{aligned}$$

which is negligible as δ is negligible. Hence, by definition, Π is correct and also 2δ-correct. ∎

Claim 1.2. The PKE scheme $\Pi = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ is IND-CPA-secure.

Proof. We prove by contradiction. Suppose to the contrary that $\Pi = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ is not IND-CPA-secure. This means there exist a PPT adversary $\mathbf{Adv}$ and a nonnegligible function $\varepsilon(\cdot)$ such that in the following experiment $\mathsf{PKE}^{\mathsf{IND\text{-}CPA}}_{\Pi}$, $\mathbf{Adv}$ wins with advantage $\varepsilon(n)$, i.e.,

$$\Pr[\mathbf{Adv} \text{ wins } \mathsf{PKE}^{\mathsf{IND\text{-}CPA}}_{\Pi}] \equiv \Pr_{\substack{((k_0, m_0), (k_1, m_1)) \leftarrow \mathbf{Adv}(\mathsf{pk}) \\ b \xleftarrow{\$} \{0,1\}}} \left[\mathbf{Adv}\left(\mathsf{Enc}^0_{\mathsf{pk}}(k_b, m_n^0), \mathsf{Enc}^0_{\mathsf{pk}}(k_n^0, m_b)\right) = b\right] = \frac{1}{2} + \varepsilon(n), \tag{1.2}$$

where the probability is taken over $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}^0(1^n)$. ∎

cont'd.

$$\begin{array}{ccc}
\mathsf{Ch} & & \mathbf{Adv} \\
(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^n) = \mathsf{KeyGen}^0(1^n) & \xrightarrow{\mathsf{pk}} & \\
b \xleftarrow{\$} \{0,1\} & \xleftarrow{((k_0,m_0),(k_1,m_1))} & \\
(ct_1, ct_2) \coloneqq \mathsf{Enc}_{\mathsf{pk}}(k_b, m_b) = \left(\mathsf{Enc}^0_{\mathsf{pk}}(k_b, m^0_n), \mathsf{Enc}^0_{\mathsf{pk}}(k^0_n, m_b)\right) & \xrightarrow{(ct_1,ct_2)} & \\
\mathbf{Adv} \text{ wins if } b' = b & \xleftarrow{b'} &
\end{array}$$

Figure 1: Experiment $\mathsf{PKE}^{\mathsf{IND\text{-}CPA}}_{\Pi}$

We construct a PPT adversary (a reduction) R that breaks the IND-CPA security of $\Pi^0 = (\mathsf{KeyGen}^0, \mathsf{Enc}^0, \mathsf{Dec}^0)$. The construction of R is provided in the following experiment $\mathsf{PKE}^{\mathsf{IND\text{-}CPA}}_{\Pi^0}$. Worth noting, R uniformly selects $i \in \{0,1\}$ and behaves differently according to i because this allows us to analyze the advantage of R by hybrid argument.

$$\begin{array}{ccccc}
\mathsf{Ch} & & R & & \mathbf{Adv} \\
(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{KeyGen}^0(1^n) & \xrightarrow{\mathsf{pk}} & i \xleftarrow{\$} \{0,1\} & \xrightarrow{\mathsf{pk}} & \\
b \xleftarrow{\$} \{0,1\} & \xleftarrow{((k_0^\star,m_0^\star),(k_1^\star,m_1^\star))} & \begin{array}{l}(k_0^\star, m_0^\star), (k_1^\star, m_1^\star) \\ \coloneqq \begin{cases} (k_0, m^0_n), (k_1, m^0_n) & \text{if } i = 0 \\ (k^0_n, m_0), (k^0_n, m_1) & \text{if } i = 1 \end{cases}\end{array} & \xleftarrow{((k_0,m_0),(k_1,m_1))} & \\
ct \coloneqq \mathsf{Enc}^0_{\mathsf{pk}}(k_b^\star, m_b^\star) & \xrightarrow{ct} & (ct_1, ct_2) \coloneqq \begin{cases} \left(ct, \mathsf{Enc}^0_{\mathsf{pk}}(k^0_n, m_1)\right) & \text{if } i = 0 \\ \left(\mathsf{Enc}^0_{\mathsf{pk}}(k_0, m^0_n), ct\right) & \text{if } i = 1 \end{cases} & \xrightarrow{(ct_1,ct_2)} & \\
R \text{ wins if } b' = b & \xleftarrow{b'} & & \xleftarrow{b'} &
\end{array}$$

Figure 2: Experiment $\mathsf{PKE}^{\mathsf{IND\text{-}CPA}}_{\Pi^0}$

Clearly R runs in PPT as $\mathbf{Adv}$ is a PPT algorithm. In the experiment $\mathsf{PKE}^{\mathsf{IND\text{-}CPA}}_{\Pi^0}$, for all $n \in \mathbb{N}$ the win rate of R is

$$\begin{aligned}
&\Pr\left[R \text{ wins } \mathsf{PKE}^{\mathsf{IND\text{-}CPA}}_{\Pi^0}\right] = \Pr_{i \xleftarrow{\$} \{0,1\}, b \xleftarrow{\$} \{0,1\}} \left[\mathbf{Adv}\left(ct_1, ct_2\right) = b\right] \\
=& \frac{1}{2} \Pr_{b \xleftarrow{\$} \{0,1\}} \left[\mathbf{Adv}\left(\mathsf{Enc}^0_{\mathsf{pk}}(k_b, m^0_n), \mathsf{Enc}^0_{\mathsf{pk}}(k^0_n, m_1)\right) = b\right] + \frac{1}{2} \Pr_{b \xleftarrow{\$} \{0,1\}} \left[\mathbf{Adv}\left(\mathsf{Enc}^0_{\mathsf{pk}}(k_0, m^0_n), \mathsf{Enc}^0_{\mathsf{pk}}(k^0_n, m_b)\right) = b\right] \\
=& \frac{1}{2}\left(\frac{1}{2}\Pr\left[\mathbf{Adv}\left(\mathsf{Enc}^0_{\mathsf{pk}}(k_1, m^0_n), \mathsf{Enc}^0_{\mathsf{pk}}(k^0_n, m_1)\right) = 1\right] + \frac{1}{2}\Pr\left[\mathbf{Adv}\left(\mathsf{Enc}^0_{\mathsf{pk}}(k_0, m^0_n), \mathsf{Enc}^0_{\mathsf{pk}}(k^0_n, m_1)\right) = 0\right]\right) \\
+& \frac{1}{2}\left(\frac{1}{2}\Pr\left[\mathbf{Adv}\left(\mathsf{Enc}^0_{\mathsf{pk}}(k_0, m^0_n), \mathsf{Enc}^0_{\mathsf{pk}}(k^0_n, m_1)\right) = 1\right] + \frac{1}{2}\Pr\left[\mathbf{Adv}\left(\mathsf{Enc}^0_{\mathsf{pk}}(k_0, m^0_n), \mathsf{Enc}^0_{\mathsf{pk}}(k^0_n, m_0)\right) = 0\right]\right) \\
=& \frac{1}{4} \cdot 1 + \frac{1}{2}\left(\frac{1}{2}\Pr\left[\mathbf{Adv}\left(\mathsf{Enc}^0_{\mathsf{pk}}(k_0, m^0_n), \mathsf{Enc}^0_{\mathsf{pk}}(k^0_n, m_0)\right) = 0\right] + \frac{1}{2}\Pr\left[\mathbf{Adv}\left(\mathsf{Enc}^0_{\mathsf{pk}}(k_1, m^0_n), \mathsf{Enc}^0_{\mathsf{pk}}(k^0_n, m_1)\right) = 1\right]\right) \\
=& \frac{1}{4} + \frac{1}{2} \Pr_{b \xleftarrow{\$} \{0,1\}} \left[\mathbf{Adv}\left(\mathsf{Enc}^0_{\mathsf{pk}}(k_b, m^0_n), \mathsf{Enc}^0_{\mathsf{pk}}(k^0_n, m_b)\right) = b\right] \\
=& \frac{1}{4} + \frac{1}{2}\left(\frac{1}{2} + \varepsilon(n)\right) \qquad \text{(by (1.2))} \\
=& \frac{1}{2} + \frac{1}{2}\varepsilon(n),
\end{aligned}$$

where all probabilities are taken over $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}^0(1^n)$, $((k_0, m_0), (k_1, m_1)) \leftarrow \mathbf{Adv}(\mathsf{pk})$.

Since the advantage $\frac{1}{2}\varepsilon(n)$ of the PPT adversary R in the experiment $\mathsf{PKE}^{\mathsf{IND\text{-}CPA}}_{\Pi^0}$ is nonnegligible, by definition, Π^0 is not IND-CPA-secure, which contradicts our assumption. ■

Now it suffices to show that the corresponding $\Pi' = (\mathsf{KeyGen}', \mathsf{Enc}', \mathsf{Dec}')$ is not IND-CCA-secure. To show this, we construct an oracle adversary $\mathbf{Adv}^{\Pi'}$ that breaks the IND-CCA security of $\Pi' = (\mathsf{KeyGen}', \mathsf{Enc}', \mathsf{Dec}')$. The construction of $\mathbf{Adv}^{\Pi'}$ is given in the following experiment $\mathsf{PKE}^{\mathsf{IND\text{-}CCA}}_{\Pi'}$, where we have expanded the definitions of $\mathsf{KeyGen}', \mathsf{Enc}', \mathsf{Dec}'$ for convenience.

> Idea. Our main idea is to replace the encrypted key k in the received ciphertext ct with another key k', sign the new, distinct ciphertext ct' with key k' to obtain a valid tag t', and then query (ct', t') to the decryption oracle $\mathsf{Dec}'_{\mathsf{sk}}(\cdot)$, where $(ct', t') \neq (ct, t)$.

Ch		$\mathbf{Adv}^{\Pi'}$
$(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^n) = \mathsf{KeyGen}^0(1^n)$	$\xrightarrow{\mathsf{pk}}$	
	$\xleftarrow{(m_0, m_1)}$	Sample $m_0, m_1 \xleftarrow{\$} \mathcal{M}_n$ where $m_0 \neq m_1$
$b \xleftarrow{\$} \{0,1\}$ $k \xleftarrow{\$} \mathcal{K}_n$ $ct = (ct_1, ct_2) \leftarrow \mathsf{Enc}_{\mathsf{pk}}(k, m_b) = \left(\mathsf{Enc}^0_{\mathsf{pk}}(k, m^0_n), \mathsf{Enc}^0_{\mathsf{pk}}(k^0_n, m_b)\right)$ $t \leftarrow \mathsf{Sign}_k(ct)$		
	$\xrightarrow{(ct=(ct_1,ct_2),t)}$	
		$k' \xleftarrow{\$} \mathcal{K}_n$ $ct' \leftarrow \left(\mathsf{Enc}^0_{\mathsf{pk}}(k', m^0_n), ct_2\right)$ $t' \leftarrow \mathsf{Sign}_{k'}(ct')$ If $(ct', t') = (ct, t)$ Then ABORT
$(k'', m') := \mathsf{Dec}_{\mathsf{sk}}(ct')$ $\widetilde{m} := \begin{cases} m' & \text{if } (k'', m') \neq \perp \wedge \mathsf{Vrfy}_{k''}(ct', t') = \mathsf{accept} \\ \perp & \text{otherwise} \end{cases}$	$\xleftarrow{(ct',t') \neq (ct,t)}$	Query $\mathsf{Dec}'_{\mathsf{sk}}(\cdot)$
	$\xrightarrow{\widetilde{m}}$	
		$b' := \begin{cases} 0 & \text{if } \widetilde{m} = m_0 \\ 1 & \text{if } \widetilde{m} = m_1 \\ \xleftarrow{\$} \{0,1\} & \text{otherwise} \end{cases}$
$\mathbf{Adv}^{\Pi'}$ wins if $b' = b$	$\xleftarrow{b'}$	

Figure 3: Experiment $\mathsf{PKE}^{\mathsf{IND\text{-}CCA}}_{\Pi'}$

Clearly $\mathbf{Adv}^{\Pi'}$ is a PPT algorithm.

Recall that $\Pi^0 = (\mathsf{KeyGen}^0, \mathsf{Enc}^0, \mathsf{Dec}^0)$ is γ-spread where $2^{-\gamma} = \mathsf{negl}(n)$, meaning that for each $n \in \mathbb{N}$ and any $(k, m) \in \mathcal{K}_n \times \mathcal{M}_n, ct \in \mathcal{C}_n$, we have

$$\Pr\left[ct = \mathsf{Enc}^0_{\mathsf{pk}}(k, m)\right] \leq 2^{-\gamma} = \mathsf{negl}(n), \tag{1.3}$$

where the probability is taken over the randomness of $\mathsf{Enc}^0_{\mathsf{pk}}$ and $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}^0(1^n)$. Using this, we show that the probability that $\mathbf{Adv}^{\Pi'}$ aborts, denoted $\Pr[\text{ABORT}]$, is negligible:

$$\Pr[\text{ABORT}] = \Pr[(ct' = t') = (ct, t)] \leq \Pr[ct' = ct] \leq \Pr\left[\mathsf{Enc}^0_{\mathsf{pk}}(k', m^0_n) = ct_1\right] \overset{(1.3)}{\leq} 2^{-\gamma} = \mathsf{negl}(n). \tag{1.4}$$

Moreover, since $\Pi = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ is 2δ-correct (which is proven by Claim 1.1) and $\delta = \mathsf{negl}(n)$, meaning that for each $n \in \mathbb{N}$ and $(k, m) \in \mathcal{K}_n \times \mathcal{M}_n$

$$\Pr\left[\mathsf{Dec}_{\mathsf{sk}}\left(\mathsf{Enc}_{\mathsf{pk}}(k, m)\right) \neq (k, m)\right] \leq 2\delta = \mathsf{negl}(n), \tag{1.5}$$

we see that the probability that ct' is correctly decrypted back to (k', m_b) is $1 - \mathsf{negl}(n)$:

$$\begin{aligned}
\Pr[(k'', m') = (k', m_b)] &= \Pr\left[\mathsf{Dec}_{\mathsf{sk}}\left(\mathsf{Enc}^0_{\mathsf{pk}}(k', m^0_n), \mathsf{Enc}^0_{\mathsf{pk}}(k^0_n, m_b)\right) = (k', m_b)\right] \\
&= \Pr\left[\mathsf{Dec}_{\mathsf{sk}}\left(\mathsf{Enc}_{\mathsf{pk}}(k', m_b)\right) = (k', m_b)\right] \\
&\overset{(1.5)}{\geq} 1 - 2\delta = 1 - \mathsf{negl}(n).
\end{aligned} \tag{1.6}$$

Also, conditioning on $(k'', m') = (k', m_b)$ (actually $k'' = k'$ suffices), the probability that $\mathsf{Vrfy}_{k''}(ct', t') = \mathsf{accept}$ (i.e. MAC tag t' is valid for message ct') is 1 by the correctness of the MAC system $\mathcal{I} = (\mathsf{Sign}, \mathsf{Vrfy})$:

$$\Pr\left[\mathsf{Vrfy}_{k''}(ct', t') = \mathsf{accept} \mid (k'', m') = (k', m_b)\right] = \Pr\left[\mathsf{Vrfy}_{k'}(ct', \mathsf{Sign}_{k'}(ct')) = \mathsf{accept}\right] = 1. \tag{1.7}$$

Putting it all together, we see that in the experiment $\mathsf{PKE}^{\text{IND-CCA}}_{\Pi'}$, since $m_0 \neq m_1$, $\mathbf{Adv}^{\Pi'}$ wins if $\widetilde{m} = m_b \wedge \neg\text{ABORT}$, i.e., $\mathbf{Adv}^{\Pi'}$ correctly obtains m_b via the decryption oracle without aborting. Therefore, for all $n \in \mathbb{N}$

$$\begin{aligned}
\Pr[\mathbf{Adv}^{\Pi'} \text{ wins } \mathsf{PKE}^{\text{IND-CCA}}_{\Pi'}] &\geq \Pr\left[\widetilde{m} = m_b \wedge \neg\text{ABORT}\right] \\
&= \Pr\left[m' = m_b \wedge (k'', m') \neq \perp \wedge \mathsf{Vrfy}_{k''}(ct', t') = \mathsf{accept} \wedge \neg\text{ABORT}\right] \\
&\geq \Pr\left[(k'', m') = (k', m_b) \wedge \mathsf{Vrfy}_{k''}(ct', t') = \mathsf{accept} \wedge \neg\text{ABORT}\right] \\
&\geq \Pr\left[(k'', m') = (k', m_b) \wedge \mathsf{Vrfy}_{k''}(ct', t') = \mathsf{accept}\right] - \Pr[\text{ABORT}] \qquad \text{(by Union Bound)} \\
&= \Pr\left[(k'', m') = (k', m_b)\right] \Pr\left[\mathsf{Vrfy}_{k''}(ct', t') = \mathsf{accept} \mid (k'', m') = (k', m_b)\right] - \Pr[\text{ABORT}] \\
&\geq (1 - \mathsf{negl}(n)) \cdot 1 - \mathsf{negl}(n) \qquad \text{(by (1.4), (1.6), (1.7))} \\
&= \frac{1}{2} + \mathsf{nonnegl}(n).
\end{aligned}$$

This implies that the advantage of $\mathbf{Adv}^{\Pi'}$ in the experiment $\mathsf{PKE}^{\text{IND-CCA}}_{\Pi'}$ is nonnegligible.

Hence, by definition, $\Pi' = (\mathsf{KeyGen}', \mathsf{Enc}', \mathsf{Dec}')$ is not IND-CCA-secure. $\circledast$

Problem 2 (Various FO transformation).

In this problem, let $\Pi = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ be a PKE scheme.

(i) Suppose that Π is δ-correct, γ-spread and OW-CPA secure, where δ and $2^{-\gamma}$ are negligible. Let $G : \mathcal{M} \to \mathcal{R}$ be a random oracle and T be the transformation defined in the class. Show that $\mathsf{T}[\Pi, G]$ is OW-CCA secure.

In the following we introduce another variant $\mathsf{U}^{\not\perp}$ of the U transformation taught in the class that converts a OW-CCA PKE scheme into an IND-CCA KEM. The transformation $\mathsf{U}^{\not\perp}$ is used in CRYSTALS-Kyber, the NIST PQC standard KEM, works similarly to the U transformation defined in the class but with "implicit rejection" of inconsistent ciphertexts. The construction is given as follows: the transformation $\mathsf{U}^{\not\perp}$ takes as input a PKE scheme $\Pi = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ with message space $\mathcal{M}$ and an RO $H : \mathcal{M} \times \mathcal{C} \to \mathcal{K}$, and outputs a KEM $\Pi_{\mathsf{KEM}^{\not\perp}} = \mathsf{U}^{\not\perp}[\Pi, H] = (\mathsf{KeyGen}^{\not\perp}, \mathsf{Encaps}, \mathsf{Decaps}^{\not\perp})$. The algorithms are defined below:

$\mathsf{KeyGen}^{\not\perp}(1^n)$	$\mathsf{Encaps}(\mathsf{pk})$	$\mathsf{Decaps}^{\not\perp}_{\mathsf{sk}'}(\mathsf{ct})$
$(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^n)$	$m \xleftarrow{\$} \mathcal{M}$	$(\mathsf{sk}, s) := \mathsf{sk}'$
$s \xleftarrow{\$} \mathcal{M}$	$\mathsf{ct} \leftarrow \mathsf{Enc}_{\mathsf{pk}}(m)$	$m' := \mathsf{Dec}_{\mathsf{sk}}(\mathsf{ct})$
$\mathsf{sk}' := (\mathsf{sk}, s)$	$k := H(m, \mathsf{ct})$	**if** $m' \neq \perp$ **then**
return $(\mathsf{pk}, \mathsf{sk}')$	**return** (k, ct)	**return** $k := H(m, \mathsf{ct})$
		else
		return $k := H(s, \mathsf{ct})$

One can see that the Encaps is the same as the one defined in the class. The $\mathsf{U}^{\not\perp}$ and U essentially differ in decapsulation: Decaps from U rejects if ct decrypts to $\perp$, whereas $\mathsf{Decaps}^{\not\perp}$ from $\mathsf{U}^{\not\perp}$ returns a pseudorandom key k.

(ii) Show that if $\Pi = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ is a OW-CCA secure PKE scheme and $H : \mathcal{M} \times \mathcal{C} \to \mathcal{K}$ is an RO, then $\Pi_{\mathsf{KEM}^{\not\perp}} = \mathsf{U}^{\not\perp}[\Pi, H]$ is IND-CCA secure.

> Remark. Actually both U and $\mathsf{U}^{\not\perp}$ can take as input a OW-CCA PKE with randomized encryption and output an IND-CCA KEM. However there are also other variants, for instances, U_m and $\mathsf{U}^{\not\perp}_m$, whose output KEMs derive key as $k = H(m)$, require that the input PKE is deterministic as the output of T transformation.

Answer (i). We prove this by contradiction. Suppose to the contrary that $\mathsf{T}[\Pi, G] = (\mathsf{KeyGen}', \mathsf{Enc}', \mathsf{Dec}')$ is not OW-CCA secure in the random oracle model. This means there exists a PPT oracle adversary $\mathbf{Adv}$ such that in the experiment PKE^{OW-CCA} (given in Figure 4) under the random oracle model, the advantage of $\mathbf{Adv}$, i.e. $\Pr[\mathbf{Adv}$ wins $\mathsf{PKE}^{OW-CCA}]$, is nonnegligible (as a function of n).

Note that Enc is randomized yet Enc$'$ is deterministic. Moreover, in the experiment PKE^{OW-CCA}, the random oracle G is simulated by lazy sampling: the challenger Ch maintains a set $\mathfrak{L}_G \subseteq \mathcal{M}_n \times \mathcal{R}_n$ of all random oracle queries made by $\mathbf{Adv}$ so far, with the convention that

$$G(m) = r \iff (m, r) \in \mathfrak{L}_G.$$

for all messages $m \in \mathcal{M}_n$ that have been queried to $G(\cdot)$. Also, since the challenger knows m^* and its corresponding randomness r^*, we can store (m^*, r^*) explicitly in the set $\mathfrak{L}_G$, which is no longer the case in the reduction R.

Ch		Adv
$(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{KeyGen}'(1^n) = \mathsf{KeyGen}(1^n)$		
$m^* \xleftarrow{\$} \mathcal{M}_n, r^* \xleftarrow{\$} \mathcal{R}_n$		
$\mathfrak{L}_G := \{(m^*, r^*)\}\ \ (\because (m,r) \in \mathfrak{L}_G \iff G(m) = r)$		
$ct^* := \mathsf{Enc}'_{\mathsf{pk}}(m^*) = \mathsf{Enc}_{\mathsf{pk}}(m^*; r^*)$	$\xrightarrow{(\mathsf{pk},ct^*)}$	
	$\xleftarrow{m_i}$	
If $\exists r$ s.t. $(m_i, r) \in \mathfrak{L}_G$ Then $r_i := r$ Else $r_i \xleftarrow{\$} \mathcal{R}_n;\ \mathfrak{L}_G := \mathfrak{L}_G \cup \{(m_i, r_i)\}$		Query $G(\cdot)$
	$\xrightarrow{r_i}$	
	$\xleftarrow{ct_i \neq ct^*}$	
$m'_i := \mathsf{Dec}'_{\mathsf{sk}}(ct_i) = \begin{cases} \mathsf{Dec}_{\mathsf{sk}}(ct_i) & \text{if } \begin{array}{l}\mathsf{Dec}_{\mathsf{sk}}(ct_i) \neq \perp \wedge \\ \mathsf{Enc}_{\mathsf{pk}}(\mathsf{Dec}_{\mathsf{sk}}(ct_i); G(\mathsf{Dec}_{\mathsf{sk}}(ct_i))) = ct_i\end{array} \\ \perp & \text{otherwise} \end{cases}$		Query $\mathsf{Dec}'_{\mathsf{sk}}(\cdot)$
where $G(\cdot)$ is the same oracle provided by Ch	$\xrightarrow{m'_i}$	
Adv wins if $m' = m^*$	$\xleftarrow{m'}$	

Figure 4: Experiment PKE^{OW-CCA}

Let $q_G = O(\mathsf{poly}(n))$ (resp. $q_{\mathsf{Dec}'} = O(\mathsf{poly}(n))$) be an upper bound of the number of issues that **Adv** queries to the random oracle G (resp. $\mathsf{Dec}'_{\mathsf{sk}}(\cdot)$). We construct a PPT reduction R that breaks the OW-CPA security of Π. The construction of R is provided in the experiment PKE^{OW-CPA} (cf. Figure 5). Worth noting, R assigns $G(m^*) \triangleq r^*$ (which is valid as both are fresh and uniform) and similarly uses lazy sampling to simulate the random oracle G: it maintains a set $\mathfrak{L}_G \subseteq \mathcal{M}_n \times \mathcal{R}_n$ such that

$$G(m) = r \iff (m, r) \in \mathfrak{L}_G \vee (m, r) = (m^*, r^*)$$

for all messages $m \in \mathcal{M}_n$ that have been queried to $G(\cdot)$, as R does not know m^*, r^* sampled by the challenger.

Ch		R		Adv
$(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^n)$				
$m^* \xleftarrow{\$} \mathcal{M}_n, r^* \xleftarrow{\$} \mathcal{R}_n$				
$ct^* := \mathsf{Enc}_{\mathsf{pk}}(m^*; r^*)$	$\xrightarrow{(\mathsf{pk},ct^*)}$	$\mathfrak{L}_G := \varnothing$	$\xrightarrow{(\mathsf{pk},ct^*)}$	
			$\xleftarrow{m_i}$	
		If $\exists r$ s.t. $(m_i, r) \in \mathfrak{L}_G$ Then $r_i := r$ Else $r_i \xleftarrow{\$} \mathcal{R}_n;\ \mathfrak{L}_G := \mathfrak{L}_G \cup \{(m_i, r_i)\}$		Query $G(\cdot)$
			$\xrightarrow{r_i}$	
			$\xleftarrow{ct_i \neq ct^*}$	
		If $\exists (m, r) \in \mathfrak{L}_G$ s.t. $\mathsf{Enc}_{\mathsf{pk}}(m; r) = ct_i$ Then $m'_i := m$ Else $m'_i := \perp$		Query $\mathsf{Dec}'_{\mathsf{sk}}(\cdot)$
			$\xrightarrow{m'_i}$	
R wins if $m'' = m^*$	$\xleftarrow{m''}$	$m'' \begin{cases} := m' & \text{with } \frac{1}{q_G+1} \text{ prob.} \\ \xleftarrow{\$} \{m \mid (m, \cdot) \in \mathfrak{L}_G\} & \text{with } \frac{q_G}{q_G+1} \text{ prob.} \end{cases}$	$\xleftarrow{m'}$	

Figure 5: Experiment PKE^{OW-CPA}

Clearly R runs in PPT since $\mathbf{Adv}$ is a PPT algorithm and the size of $\mathfrak{L}_G$ is at most the number of queries made by $\mathbf{Adv}$ to G, which is bounded above by $q_G = O(\mathsf{poly}(n))$.

WLOG, we assume $\mathbf{Adv}$ always queries distinct m_i (resp. ct_i) to $G(\cdot)$ (resp. $\mathsf{Dec}'_{\mathsf{sk}}(\cdot)$) as we may otherwise construct another $\mathbf{Adv}'$ that internally records the answers and answers itself on repeated queries. To see why in the experiment PKE^{OW-CPA}, believing in the output of $\mathbf{Adv}$ with probability only $\frac{1}{q_G+1}$ while uniformly sampling a previously queried message from the "log" $\mathfrak{L}_G$ with probability $\frac{q_G}{q_G+1}$ in the last step guarantees R a nonnegligible advantage, we make the following observations.

1. In the experiment PKE^{OW-CPA}, under the following three conditions will both oracles, namely $G(\cdot)$ and $\mathsf{Dec}'_{\mathsf{sk}}(\cdot)$, simulated by R here behave identically to (i.e., output values distributed identically to) the real oracles in Figure 4:

Event I There is no query ct_i to $\mathsf{Dec}'_{\mathsf{sk}}(\cdot)$ such that (1) $m''_i \neq \perp$ (i.e. $m''_i \in \mathcal{M}_n$), (2) $m''_i \notin \{m \mid (m, \cdot) \in \mathfrak{L}_G\}$ (i.e. m''_i has never been queried to $G(\cdot)$ before), and (3) $\mathsf{Enc}_{\mathsf{pk}}(m''_i; G(m''_i)) = ct_i$, where we define $m'' \triangleq \mathsf{Dec}_{\mathsf{sk}}(ct_i)$.

Event II Every (m_i, r_i) is not *problematic*, where $m_1, m_2, \ldots$ are all (distinct) queries to $G(\cdot)$, $r_1, r_2, \ldots \xleftarrow{\$} \mathcal{R}_n$ are the corresponding randomness stored in $\mathfrak{L}_G$, and we call a message-randomness pair $(m, r) \in \mathcal{M}_n \times \mathcal{R}_n$ *problematic* if it triggers a correctness error in Π, i.e., $\mathsf{Dec}_{\mathsf{sk}}(\mathsf{Enc}_{\mathsf{pk}}(m; r)) \neq m$.

Event III m^* has never been queried to $G(\cdot)$.

It's straightforward to see that given III (i.e., m^* has never been queried to $G(\cdot)$), R perfectly simulates $G(\cdot)$, as $m_i = m^*$ is the only query that will trigger the discrepancy between outputs of the simulated $G(\cdot)$ and real $G(\cdot)$: on $m_i = m^*$, the simulated one samples a fresh $r_i \xleftarrow{\$} \mathcal{R}_n$ whilst the real one returns r^*.
Next, we argue that conditioning on I, II, and III, the simulated and real decryption oracles $\mathsf{Dec}'_{\mathsf{sk}}(\cdot)$ output identical values for every query, by considering a single query ct_i to $\mathsf{Dec}'_{\mathsf{sk}}(\cdot)$ and define $m''_i \triangleq \mathsf{Dec}_{\mathsf{sk}}(ct_i)$:

- If m''_i has never been queried to $G(\cdot)$ before, then, on the one hand, we see that in Figure 4, $m''_i \neq m^*$ by III and thus (2) must hold, so either (1) or (3) must fail by I, which implies that in the real decryption oracle, either the decryption step or re-encryption step fails, so it will output $\perp$. On the other hand, suppose in Figure 5 $\exists (m, r) \in \mathfrak{L}_G$ such that $\mathsf{Enc}_{\mathsf{pk}}(m; r) = ct_i$, then we would have $m''_i = \mathsf{Dec}_{\mathsf{sk}}(ct_i) = \mathsf{Dec}_{\mathsf{sk}}(\mathsf{Enc}_{\mathsf{pk}}(m; r)) = m \in \mathcal{M}_n$, where the last equation follows from II (which tells us (m, r) is not *problematic*). This would imply $m''_i = m$ has been queried to $G(\cdot)$ before, which is a contradiction, so the simulated decryption oracle will output $\perp$. Hence we know both decryption oracles output $\perp$.
- Otherwise, m''_i has been queried to $G(\cdot)$ before, and by III we must have $m''_i \neq m^*$. This implies that in both experiments, $(m''_i, r_i) \in \mathfrak{L}_G$ where $G(m''_i) \triangleq r_i \xleftarrow{\$} R_n$, which implies by II that $\mathsf{Dec}_{\mathsf{sk}}(\mathsf{Enc}_{\mathsf{pk}}(m''_i; r_i)) = m''_i$. On the one hand, we see that in Figure 4, since $m''_i \neq \perp$ and $\mathsf{Enc}_{\mathsf{pk}}(m''_i; G(m''_i)) = \mathsf{Enc}_{\mathsf{pk}}(m''_i; r_i)$, the real decryption oracle outputs m''_i if $\mathsf{Enc}_{\mathsf{pk}}(m''_i; r_i) = ct_i$ and $\perp$ otherwise. On the other hand, in Figure 5, the simulated decryption oracle also outputs m''_i if $\mathsf{Enc}_{\mathsf{pk}}(m''_i; r_i) = ct_i$ and $\perp$ otherwise. Hence we know both decryption oracles output the same value.

We conclude that if in the last step R chooses to believe in the answer returned by $\mathbf{Adv}$, then

$$\begin{aligned}
&\Pr[R \text{ wins } \mathsf{PKE}^{OW-CPA} \mid m'' := m'] \\
\geq &\Pr[m' = m^*] \\
\geq &\Pr[m' = m^* \wedge I \wedge II \wedge III] \\
= &\Pr[\mathbf{Adv} \text{ wins } \mathsf{PKE}^{OW-CCA} \wedge I \wedge II \wedge III] \\
&\qquad \text{(since in both experiemnts, inputs to } \mathbf{Adv} \text{ are identically distributed} \\
&\qquad\qquad \text{and each oracle } (G(\cdot) \text{ and } \mathsf{Dec}'_{\mathsf{sk}}(\cdot)) \text{ behaves identically for every query)} \\
\geq &\Pr[\mathbf{Adv} \text{ wins } \mathsf{PKE}^{OW-CCA}] - \Pr[\neg I] - \Pr[\neg II] - \Pr[\neg III]. \qquad \text{(by Union Bound)}
\end{aligned}$$

That is,

$$\Pr[\mathbf{Adv} \text{ wins } \mathsf{PKE}^{OW-CCA}] \leq \Pr[\neg I] + \Pr[\neg II] + \Pr[\neg III] + \Pr[R \text{ wins } \mathsf{PKE}^{OW-CPA} \mid m'' := m']. \tag{2.1}$$

2. Note that there are at most $q_{\mathsf{Dec}'}$ queries to the decryption oracle $\mathsf{Dec}'_{\mathsf{sk}}(\cdot)$. Hence, by γ-spreadness of Π, the probability that Event I does not happen is actually negligible:

$$\begin{aligned}
\Pr[\neg I] &= \Pr\left[\bigvee_{i=1}^{q_{\mathsf{Dec}'}} (m''_i \in \mathcal{M}_n \wedge m''_i \notin \{m \mid (m, \cdot) \in \mathfrak{L}_G\} \wedge \mathsf{Enc}_{\mathsf{pk}}(m''_i; G(m''_i)) = ct_i)\right] \qquad (\text{recall } m''_i \triangleq \mathsf{Dec}_{\mathsf{sk}}(ct_i)) \\
&\leq \sum_{i=1}^{q_{\mathsf{Dec}'}} \Pr\left[m''_i \in \mathcal{M}_n \wedge m''_i \notin \{m \mid (m, \cdot) \in \mathfrak{L}_G\} \wedge \mathsf{Enc}_{\mathsf{pk}}(m''_i; G(m''_i)) = ct_i\right] \qquad \text{(by Union Bound)} \\
&\leq \sum_{i=1}^{q_{\mathsf{Dec}'}} \Pr\left[\mathsf{Enc}_{\mathsf{pk}}(m''_i; G(m''_i)) = ct_i \,\middle|\, m''_i \in \mathcal{M}_n \wedge m''_i \notin \{m \mid (m, \cdot) \in \mathfrak{L}_G\}\right] \\
&= \sum_{i=1}^{q_{\mathsf{Dec}'}} \Pr_{r \xleftarrow{\$} \mathcal{R}_n}\left[\mathsf{Enc}_{\mathsf{pk}}(m''_i; r) = ct_i \,\middle|\, m''_i \in \mathcal{M}_n \wedge m''_i \notin \{m \mid (m, \cdot) \in \mathfrak{L}_G\}\right] \\
&\qquad (\text{since } m''_i \in \mathcal{M}_n \text{has never been queried to } G(\cdot) \text{ before, } G(m''_i) \in \mathcal{R}_n \text{ is uniform and fresh}) \\
&\leq \sum_{i=1}^{q_{\mathsf{Dec}'}} 2^{-\gamma} \qquad (\text{since } \Pi \text{ is } \gamma\text{-spread, i.e., } \Pr_{r \xleftarrow{\$} \mathcal{R}_n}[ct = \mathsf{Enc}_{\mathsf{pk}}(m; r)] \leq 2^{-\gamma} \ \ \forall m \in \mathcal{M}_n, ct \in \mathcal{C}_n) \\
&= q_{\mathsf{Dec}'} \cdot 2^{-\gamma}.
\end{aligned} \tag{2.2}$$

(Note that all probabilities above are taken over $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^n)$.)

3. Note that there are at most q_G queries to the random oracle $G(\cdot)$. Hence, by δ-correctness of Π, the probability that Event II does not happen is actually negligible:

$$\begin{aligned}
\Pr[\neg II] &= \Pr_{r_1, \ldots, r_{q_G} \xleftarrow{\$} \mathcal{R}_n}\left[\bigvee_{i=1}^{q_G} \mathsf{Dec}_{\mathsf{sk}}(\mathsf{Enc}_{\mathsf{pk}}(m_i; r_i)) \neq m_i\right] \qquad (\text{since } \{m_i\} \text{ are distinct, every } r_i \text{ is fresh}) \\
&\leq \sum_{i=1}^{q_G} \Pr_{r_i \xleftarrow{\$} \mathcal{R}_n}[\mathsf{Dec}_{\mathsf{sk}}(\mathsf{Enc}_{\mathsf{pk}}(m_i; r_i)) \neq m_i] \qquad \text{(by Union Bound)} \\
&\leq \sum_{i=1}^{q_G} \delta \qquad (\text{since } \Pi \text{ is } \delta\text{-correct, i.e., } \Pr_{r \xleftarrow{\$} \mathcal{R}_n}[\mathsf{Dec}_{\mathsf{sk}}(\mathsf{Enc}_{\mathsf{pk}}(m; r)) \neq m] \leq \delta \ \ \forall m \in \mathcal{M}_n) \\
&= q_G \cdot \delta.
\end{aligned} \tag{2.3}$$

(Note that all probabilities above are taken over $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^n)$.)

4. We do not proceed to bound $\Pr[\neg III]$. Instead, observe that if Event III fails in the experiment PKE^{OW-CPA}, which means that $\mathbf{Adv}$ has queried m^* to $G(\cdot)$, then m^* must be listed in the set $\mathfrak{L}_G$, i.e., $m^* \in \{m \mid (m,\cdot) \in \mathfrak{L}_G\}$, after $\mathbf{Adv}$ returns m' to R. Therefore, if in the last step R chooses to uniformly select a previously queried message from $\mathfrak{L}_G$, then the probability that R selects the correct answer m^* is bounded below by:

$$\begin{aligned}
\Pr[R \text{ wins } \mathsf{PKE}^{OW-CPA} \mid m'' \xleftarrow{\$} \{m \mid (m,\cdot) \in \mathfrak{L}_G\}] &= \Pr_{m'' \xleftarrow{\$} \{m \mid (m,\cdot)\in\mathfrak{L}_G\}}[m'' = m^*] \\
&\geq \Pr_{m'' \xleftarrow{\$} \{m \mid (m,\cdot)\in\mathfrak{L}_G\}}[m'' = m^* \wedge \neg III] \\
&\geq \Pr[\neg III] \Pr_{m'' \xleftarrow{\$} \{m \mid (m,\cdot)\in\mathfrak{L}_G\}}[m'' = m^* \mid \neg III] \\
&\geq \Pr[\neg III] \cdot \frac{1}{|\mathfrak{L}_G|} \qquad (\because (m^*,\cdot) \text{ must be in } \mathfrak{L}_G) \\
&\geq \Pr[\neg III] \cdot \frac{1}{q_G} \qquad (\because |\mathfrak{L}_G| \leq q_G).
\end{aligned}$$

Hence,

$$\Pr[\neg III] \leq q_G \cdot \Pr[R \text{ wins } \mathsf{PKE}^{OW-CPA} \mid m'' \xleftarrow{\$} \{m \mid (m,\cdot) \in \mathfrak{L}_G\}]. \tag{2.4}$$

Putting it all together, we have

$$\begin{aligned}
\Pr[\mathbf{Adv} \text{ wins } \mathsf{PKE}^{OW-CCA}] &\leq \Pr[\neg I] + \Pr[\neg II] + \Pr[\neg III] \\
&\quad + \Pr[R \text{ wins } \mathsf{PKE}^{OW-CPA} \mid m'' := m'] \qquad \text{(by (2.1))} \\
&\leq q_{\mathsf{Dec}'} \cdot 2^{-\gamma} + q_G \cdot \delta \\
&\quad + q_G \cdot \Pr[R \text{ wins } \mathsf{PKE}^{OW-CPA} \mid m'' \xleftarrow{\$} \{m \mid (m,\cdot) \in \mathfrak{L}_G\}] \\
&\quad + \Pr[R \text{ wins } \mathsf{PKE}^{OW-CPA} \mid m'' := m'] \qquad \text{(by (2.2), (2.3), (2.4))} \\
&= q_{\mathsf{Dec}'} \cdot 2^{-\gamma} + q_G \cdot \delta + (q_G + 1) \\
&\quad \Big(\frac{q_G}{q_G+1} \cdot \Pr[R \text{ wins } \mathsf{PKE}^{OW-CPA} \mid m'' \xleftarrow{\$} \{m \mid (m,\cdot) \in \mathfrak{L}_G\}] \\
&\quad + \frac{1}{q_G+1} \cdot \Pr[R \text{ wins } \mathsf{PKE}^{OW-CPA} \mid m'' := m'] \Big) \\
&= q_{\mathsf{Dec}'} \cdot 2^{-\gamma} + q_G \cdot \delta + (q_G + 1) \Pr[R \text{ wins } \mathsf{PKE}^{OW-CPA}], \qquad \text{(by construction of } R)
\end{aligned}$$

which gives us

$$\begin{aligned}
\Pr[R \text{ wins } \mathsf{PKE}^{OW-CPA}] &\geq \frac{1}{q_G+1} \left(\Pr[\mathbf{Adv} \text{ wins } \mathsf{PKE}^{OW-CCA}] - q_{\mathsf{Dec}'} \cdot 2^{-\gamma} - q_G \cdot \delta \right) \\
&= \underbrace{\frac{1}{O(\mathsf{poly}(n))}}_{=\mathsf{nonnegl}(n)} \underbrace{\left(\mathsf{nonnegl}(n) - \underbrace{O(\mathsf{poly}(n))\mathsf{negl}(n)}_{=\mathsf{negl}(n)} - \underbrace{O(\mathsf{poly}(n))\mathsf{negl}(n)}_{=\mathsf{negl}(n)} \right)}_{=\mathsf{nonnegl}(n)} \qquad \text{(by assumption)} \\
&= \mathsf{nonnegl}(n).
\end{aligned}$$

This means that R has nonnegligible advantage in the experiment PKE^{OW-CPA}.

Hence, by definition, Π is not OW-CPA secure, contradicting our assumption. ⊛

Answer (ii). We prove this contradiction. Suppose to the contrary that the KEM scheme $\Pi_{\mathsf{KEM}^{\not\perp}} = \mathsf{U}^{\not\perp}[\Pi, H] = (\mathsf{KeyGen}^{\not\perp}, \mathsf{Encaps}, \mathsf{Decaps}^{\not\perp})$ is not IND-CCA-secure in the random oracle model. This means that there exist a PPT oracle adversary $\mathbf{Adv}$ and a nonnegligible function $\varepsilon(\cdot)$ such that in the experiment $\mathsf{KEM}^{\mathsf{IND-CCA}}$ (cf. Figure 6) under the random oracle model, $\mathbf{Adv}$ wins with advantage $\varepsilon(n)$, i.e.,

$$\Pr\left[\mathbf{Adv} \text{ wins } \mathsf{KEM}^{\mathsf{IND-CCA}}\right] = \frac{1}{2} + \varepsilon(n) \tag{2.5}$$

for all $n \in \mathbb{N}$.

Worth noting, in the experiment $\mathsf{KEM}^{\mathsf{IND-CCA}}$, we have expanded the definitions of $\mathsf{KeyGen}^{\not\perp}, \mathsf{Encaps}, \mathsf{Decaps}^{\not\perp}$ for convenience, and applied lazy sampling to simulate the random oracle H: the challenger Ch maintains a set $\mathfrak{L}_H \subseteq (\mathcal{M}_n \times \mathcal{C}_n) \times \mathcal{K}_n$ of all random oracle queries made by $\mathbf{Adv}$ so far, with the convention that

$$H(m, \mathsf{ct}) = k \iff ((m, \mathsf{ct}), k) \in \mathfrak{L}_H. \tag{2.6}$$

for all message-ciphertext pairs $(m, \mathsf{ct}) \in \mathcal{M}_n \times \mathcal{C}_n$ that have been queried to $H(\cdot)$. Also, since the challenger knows m^*, ct^*, and its corresponding pseudorandom key k_0, we can store $((m^*, \mathsf{ct}^*), k_0)$ explicitly in the set $\mathfrak{L}_H$, which is no longer the case in the reduction R as shown later.

Ch		$\mathbf{Adv}^{H(\cdot),\mathsf{Decaps}^{\not\perp}_{\mathsf{sk}'}(\cdot)}$
$(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^n)$; $s \xleftarrow{\$} \mathcal{M}_n$; $\mathsf{sk}' := (\mathsf{sk}, s)$ $m^* \xleftarrow{\$} \mathcal{M}_n$; $\mathsf{ct}^* \leftarrow \mathsf{Enc}_{\mathsf{pk}}(m)$ $k_0 \xleftarrow{\$} \mathcal{K}_n$; $\mathfrak{L}_H := \{((m^*, \mathsf{ct}^*), k_0)\}$ $b \xleftarrow{\$} \{0,1\}$ $k_1 \xleftarrow{\$} \mathcal{K}_n$	$\xrightarrow{(\mathsf{pk}, \mathsf{ct}^*, k_b)}$	
	$\xleftarrow{m_i, \mathsf{ct}_i}$	
If $\exists k$ s.t. $((m_i, \mathsf{ct}_i), k) \in \mathfrak{L}_H$ **Then** $k_i := k$ **Else** $k_i \xleftarrow{\$} \mathcal{K}_n$; $\mathfrak{L}_H := \mathfrak{L}_H \cup \{((m_i, \mathsf{ct}_i), k_i)\}$		Query $H(\cdot)$
	$\xrightarrow{k_i}$	
	$\xleftarrow{\mathsf{ct}_i \neq \mathsf{ct}^*}$	
$m_i := \mathsf{Dec}_{\mathsf{sk}}(\mathsf{ct}_i)$ **If** $m_i \neq \perp$ **Then** $k_i := H(m_i, \mathsf{ct}_i)$ **Else** $k_i := H(s, \mathsf{ct}_i)$ where $H(\cdot)$ is the same oracle provided by Ch		Query $\mathsf{Decaps}^{\not\perp}_{\mathsf{sk}'}(\cdot)$
	$\xrightarrow{k_i}$	
$\mathbf{Adv}$ wins if $b' = b$	$\xleftarrow{b'}$	

Figure 6: Experiment $\mathsf{KEM}^{\mathsf{IND-CCA}}$

Let $q_H = O(\mathsf{poly}(n))$ (resp. $q_{\mathsf{Decaps}^{\not\perp}} = O(\mathsf{poly}(n))$) be an upper bound of the number of issues that $\mathbf{Adv}$ queries to the random oracle H (resp. $\mathsf{Decaps}^{\not\perp}_{\mathsf{sk}'}(\cdot)$). We construct a PPT reduction R that breaks the OW-CCA security of the PKE scheme $\Pi = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$. The construction of R is provided in the experiment $\mathsf{PKE}^{\mathsf{OW-CCA}}$ (cf. Figure 7). Likewise, R uses lazy sampling to simulate the random oracle H: it maintains a set $\mathfrak{L}_H \subseteq (\mathcal{M}_n \times \mathcal{C}_n) \times \mathcal{K}_n$ such that

$$H(m, \mathsf{ct}) = k \iff ((m, \mathsf{ct}), k) \in \mathfrak{L}_H \vee ((m, \mathsf{ct}), k) = ((m^*, \mathsf{ct}^*), k_0), \tag{2.7}$$

for all message-ciphertext pairs $(m, \mathsf{ct}) \in \mathcal{M}_n \times \mathcal{C}_n$ that have been queried to $H(\cdot)$. Worth noting, R samples $k_0 \triangleq H(m^*, \mathsf{ct}^*) \xleftarrow{\$} \mathcal{K}_n$ without saving it to $\mathfrak{L}_H$ because R does not know m^*, which is sampled by the challenger. (Note that R can indeed sample $H(m^*, \mathsf{ct}^*)$ as it has not been queried before and is thus fresh and uniform, albeit not knowing m^*.)

Ch		$R^{\mathsf{Dec_{sk}}(\cdot)}$		$\mathbf{Adv}^{H(\cdot),\mathsf{Decaps}^{\not\perp}_{\mathsf{sk}'}}$
$(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^n)$				
$m^* \xleftarrow{\$} \mathcal{M}_n$				
$\mathsf{ct}^* \leftarrow \mathsf{Enc_{pk}}(m^*)$	$\xrightarrow{(\mathsf{pk}, ct^*)}$	$\mathfrak{L}_H := \varnothing$		
		$s \xleftarrow{\$} \mathcal{M}_n$; $\mathsf{sk}' := (\mathsf{sk}, s)$		
		$k_0 \triangleq H(m^*, \mathsf{ct}^*) \xleftarrow{\$} \mathcal{K}_n$		
		$b \xleftarrow{\$} \{0,1\}$		
		$k_1 \xleftarrow{\$} \mathcal{K}_n$	$\xrightarrow{(\mathsf{pk}, \mathsf{ct}^*, k_b)}$	
			$\xleftarrow{m_i, \mathsf{ct}_i}$	
		If $\exists k$ s.t. $((m_i, \mathsf{ct}_i), k) \in \mathfrak{L}_H$		
		Then $k_i := k$		Query $H(\cdot)$
		Else $k_i \xleftarrow{\$} \mathcal{K}_n$;		
		$\mathfrak{L}_H := \mathfrak{L}_H \cup \{((m_i, \mathsf{ct}_i), k_i)\}$	$\xrightarrow{k_i}$	
			$\xleftarrow{\mathsf{ct}_i \neq \mathsf{ct}^*}$	
	$\xleftrightarrow{\text{Query } \mathsf{Dec_{sk}}(\mathsf{ct}_i \neq \mathsf{ct}^*)}$	$m_i := \mathsf{Dec_{sk}}(\mathsf{ct}_i)$		
		If $m_i \neq \perp$		
		Then $k_i := H(m_i, \mathsf{ct}_i)$		Query $\mathsf{Decaps}^{\not\perp}_{\mathsf{sk}'}$
		Else $k_i := H(s, \mathsf{ct}_i)$		
		where $H(\cdot)$ is the oracle provided by R		
		$\mathsf{Dec_{sk}}(\cdot)$ is the oracle provided by Ch	$\xrightarrow{k_i}$	
R wins if $m' = m^*$	$\xleftarrow{m'}$	$m' \xleftarrow{\$} \{m \mid ((m, \cdot), \cdot) \in \mathfrak{L}_H\}$	$\xleftarrow{b'}$	

Figure 7: Experiment $\mathsf{PKE}^{\mathsf{OW-CCA}}$

Clearly R runs in PPT since $\mathbf{Adv}$ is a PPT algorithm and the size of $\mathfrak{L}_H$ is at most the number of queries to $H(\cdot)$, which is bounded above by $q_H + q_{\mathsf{Decaps}^{\not\perp}} = O(\mathsf{poly}(n))$.

WLOG, we assume $\mathbf{Adv}$ always queries distinct (m_i, k_i) (resp. ct_i) to $H(\cdot)$ (resp. $\mathsf{Decaps}^{\not\perp}_{\mathsf{sk}'}(\cdot)$) as we may otherwise construct another $\mathbf{Adv}'$ that internally records the answers and answers itself on repeated queries.

Define $\mathsf{QUERY}^{\mathsf{PKE}}$ and $\mathsf{QUERY}^{\mathsf{KEM}}$ as the events on which $\mathbf{Adv}$ has ever queried (m^*, ct^*) to the oracle $H(\cdot)$ in the experiments $\mathsf{PKE}^{\mathsf{OW-CCA}}$ and $\mathsf{KEM}^{\mathsf{IND-CCA}}$, respectively. It's straightforward to see that in the experiment $\mathsf{PKE}^{\mathsf{OW-CCA}}$, before (m^*, ct^*) is ever queried by $\mathbf{Adv}$ to $H(\cdot)$, both oracles simulated by R, namely $H(\cdot)$ and $\mathsf{Decaps}^{\not\perp}_{\mathsf{sk}'}(\cdot)$, have been behaving exactly the same as the real oracles in Figure 6 (i.e., output values distributed identically to those from the real ones). We argue as follows:

- Comparing (2.6) and (2.7), we may observe that the only input that can trigger the behavioral discrepancy between oracles $H(\cdot)$ in Figure 6 (real) and in Figure 7 (simulated) is (m^*, ct^*): on $(m_i, \mathsf{ct}_i) = (m^*, \mathsf{ct}^*)$, the simulated one samples a fresh $k_i \xleftarrow{\$} \mathcal{K}_n$ whilst the real one returns k_0. Also, the oracle $\mathsf{Decaps}^{\not\perp}_{\mathsf{sk}'}(\cdot)$ will never make a query (m^*, ct^*) to $H(\cdot)$ because we know the query $\mathsf{ct}_i \neq \mathsf{ct}^*$. Therefore, before (m^*, ct^*) is ever queried by $\mathbf{Adv}$ to $H(\cdot)$, R has been perfectly simulating $H(\cdot)$.

- In the oracle $\mathsf{Decaps}^{\not\perp}_{\mathsf{sk}'}(\cdot)$ simulated by R, since $\mathsf{ct}_i \neq \mathsf{ct}^*$, the decryption oracle $\mathsf{Dec}_{\mathsf{sk}}(\cdot)$ simulated by the challenger always outputs the same values as the real decryption function $\mathsf{Dec}_{\mathsf{sk}}$ in Figure 6 (as R is only not allowed to query ct^*). Combining with the previous observation that $H(\cdot)$ has been perfectly simulated, we see that R has also been perfectly simulating $\mathsf{Decaps}^{\not\perp}_{\mathsf{sk}'}(\cdot)$. That is, the simulated $\mathsf{Decaps}^{\not\perp}_{\mathsf{sk}'}(\cdot)$ has been behaving the same as the real one in Figure 6, before (m^*, ct^*) is ever queried by $\mathbf{Adv}$ to $H(\cdot)$.

We conclude that all inputs to $\mathbf{Adv}$ and outputs from the oracles $H(\cdot)$ and $\mathsf{Decaps}^{\not\perp}_{\mathsf{sk}'}(\cdot)$ have been distributed identically to those in $\mathsf{KEM}^{\mathsf{IND-CCA}}$ before (m^*, ct^*) is ever queried by $\mathbf{Adv}$ to $H(\cdot)$. Hence we have

$$\Pr\left[\mathsf{QUERY}^{\mathsf{PKE}}\right] = \Pr\left[\mathsf{QUERY}^{\mathsf{KEM}}\right]. \tag{2.8}$$

Now observe in the experiment $\mathsf{KEM}^{\mathsf{IND-CCA}}$ that since k_0 and k_1 are both uniformly sampled, k_b is in fact independent of $b \in \{0,1\}$. Moreover, if $\neg\mathsf{QUERY}^{\mathsf{KEM}}$ (i.e., $\mathbf{Adv}$ has never queried (m^*, ct^*)), then all outputs of $H(\cdot)$ must be independent of k_0. Also, as $\mathsf{Decaps}^{\not\perp}_{\mathsf{sk}'}(\cdot)$ never queries $(\cdot, \mathsf{ct}^*)$ to $H(\cdot)$, all outputs of $\mathsf{Decaps}^{\not\perp}_{\mathsf{sk}'}(\cdot)$ must also be independent of k_0. Independence from k_1 is immediate as well. These together imply that k_b, together with all the other random variables in the *view* of $\mathbf{Adv}$ (e.g., $\mathsf{pk}, \mathsf{ct}^*$, outputs of $H(\cdot)$, outputs of $\mathsf{Decaps}^{\not\perp}_{\mathsf{sk}'}(\cdot)$, etc.), is independent of k_0 and k_1. Hence, the view of $\mathbf{Adv}$ and thus its output b' are independent of $b \in \{0,1\}$. As b is uniform, we see that $\mathbf{Adv}$ has exactly $\frac{1}{2}$ probability to correctly guess b, that is,

$$\Pr\left[\mathbf{Adv} \text{ wins } \mathsf{KEM}^{\mathsf{IND-CCA}} \,\middle|\, \neg\mathsf{QUERY}^{\mathsf{KEM}}\right] = \Pr_{b \xleftarrow{\$} \{0,1\}}\left[\mathbf{Adv}^{H(\cdot), \mathsf{Decaps}^{\not\perp}_{\mathsf{sk}'}(\cdot)}(\mathsf{pk}, \mathsf{ct}^*, k_b) = b \,\middle|\, \neg\mathsf{QUERY}^{\mathsf{KEM}}\right] = \frac{1}{2}. \tag{2.9}$$

From this we reason that

$$\begin{aligned}
\frac{1}{2} + \varepsilon(n) &= \Pr\left[\mathbf{Adv} \text{ wins } \mathsf{KEM}^{\mathsf{IND-CCA}}\right] && \text{(by (2.5))}\\
&= \Pr\left[\mathsf{QUERY}^{\mathsf{KEM}}\right] \Pr\left[\mathbf{Adv} \text{ wins } \mathsf{KEM}^{\mathsf{IND-CCA}} \,\middle|\, \mathsf{QUERY}^{\mathsf{KEM}}\right]\\
&\qquad + \Pr\left[\neg\mathsf{QUERY}^{\mathsf{KEM}}\right] \Pr\left[\mathbf{Adv} \text{ wins } \mathsf{KEM}^{\mathsf{IND-CCA}} \,\middle|\, \neg\mathsf{QUERY}^{\mathsf{KEM}}\right]\\
&\leq \Pr\left[\mathsf{QUERY}^{\mathsf{KEM}}\right] \cdot 1 + \left(1 - \Pr\left[\mathsf{QUERY}^{\mathsf{KEM}}\right]\right) \cdot \frac{1}{2} && \text{(by (2.9))}\\
&= \frac{1}{2} + \frac{1}{2} \Pr\left[\mathsf{QUERY}^{\mathsf{KEM}}\right]\\
&= \frac{1}{2} + \frac{1}{2} \Pr\left[\mathsf{QUERY}^{\mathsf{PKE}}\right], && \text{(by (2.8))}
\end{aligned}$$

which gives us

$$\Pr\left[\mathsf{QUERY}^{\mathsf{PKE}}\right] \geq 2\varepsilon(n). \tag{2.10}$$

Lastly, observe in the experiment $\mathsf{PKE}^{\mathsf{OW-CCA}}$ that given $\mathsf{QUERY}^{\mathsf{PKE}}$ (i.e., $\mathbf{Adv}$ has queried (m^*, ct^*)), then, as $m^* \in \{m \mid ((m,\cdot),\cdot) \in \mathfrak{L}_H\}$, the probability that R selects a correct $m' = m^*$ is at least

$$\Pr_{m' \xleftarrow{\$} \{m \mid ((m,\cdot),\cdot) \in \mathfrak{L}_H\}}\left[m' = m^* \,\middle|\, \mathsf{QUERY}^{\mathsf{PKE}}\right] \geq \frac{1}{|\mathfrak{L}_H|} \geq \frac{1}{q_H + q_{\mathsf{Decaps}^{\not\perp}}} = \frac{1}{O(\mathsf{poly}(n))} = \mathsf{nonnegl}(n). \tag{2.11}$$

(Recall that the size of $\mathfrak{L}_H$ is at most the number of queries to $H(\cdot)$, which is bounded above by $q_H + q_{\mathsf{Decaps}^{\not\perp}}$.) Hence, the win rate of R in $\mathsf{PKE}^{\mathsf{OW-CCA}}$ is bounded below by

$$\begin{aligned}
\Pr[R \text{ wins } \mathsf{PKE}^{\mathsf{OW-CCA}}] &= \Pr[m' = m^*] \\
&\geq \Pr\left[m' = m^* \wedge \mathsf{QUERY}^{\mathsf{PKE}}\right] \\
&= \Pr\left[\mathsf{QUERY}^{\mathsf{PKE}}\right] \Pr\left[m' = m^* \;\middle|\; \mathsf{QUERY}^{\mathsf{PKE}}\right] \\
&\geq (2\varepsilon(n)) \cdot \mathsf{nonnegl}(n) && \text{(by (2.10) and (2.11))} \\
&= 2\mathsf{nonnegl}(n) \cdot \mathsf{nonnegl}(n) && \text{(by assumption)} \\
&= \mathsf{nonnegl}(n).
\end{aligned}$$

This means that R has nonnegligible advantage in the experiment $\mathsf{PKE}^{\mathsf{OW-CCA}}$.

Hence, by definition, the PKE scheme $\Pi = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ is not OW-CCA secure, contradicting our assumption. ⊛

Problem 3 (Bit commitment in the RO model).

In this problem, we will present a concrete definition of non-interactive commitment schemes in the RO model and a simple construction, and you are asked to prove its security.

Let $n \in \mathbb{N}$ be the security parameter and $\mathsf{RO} : \{0,1\}^{n+1} \to \{0,1\}^{3n}$ be a random oracle. Consider the following simple construction $\mathsf{Com}^{\mathsf{RO}}(b; r) \triangleq \mathsf{RO}(b \parallel r)$. Show that this simple construction satisfies the following binding and hiding properties.

(i) *Hiding*: Consider the following hiding security game:

- The challenger samples $b \xleftarrow{\$} \{0,1\}$ and $r \xleftarrow{\$} \{0,1\}^n$ uniformly at random, computes and sends $\mathsf{com} = \mathsf{Com}^{\mathsf{RO}}(b; r)$ to the adversary.
- Upon receiving com, the adversary outputs a guess $b' \leftarrow \mathbf{Adv}^{\mathsf{RO}}(\mathsf{com})$.
- The adversary wins if $b' = b$.

Show that for any oracle adversary $\mathbf{Adv}^{\mathsf{RO}}$ that makes at most q queries,

$$\Pr[\mathbf{Adv}^{\mathsf{RO}} \text{ wins}] \leq \frac{1}{2} + \frac{2q}{2^n}.$$

(ii) *Binding*: Show that for any oracle adversary $\mathbf{Adv}^{\mathsf{RO}}$ that makes at most q queries,

$$\Pr\left[\mathsf{com} = \mathsf{Com}^{\mathsf{RO}}(0; r_0) = \mathsf{Com}^{\mathsf{RO}}(1, r_1) : (\mathsf{com}, r_0, r_1) \leftarrow \mathbf{Adv}^{\mathsf{RO}}()\right] \leq \frac{1}{2^{3n}} + \frac{q^2}{2^{3n}}.$$

(iii) In HW2, we also see a construction of a bit commitment scheme from PRG. They are the same primitive, but defined in different models (one in the plain model and the other in the random oracle model). This is common in theoretical cryptography. In this problem, you are asked to compare two of them.

(a) Compare the syntax, security, and hiding definitions.

(b) Compare how both constructions achieve the hiding and binding properties.

(c) Discuss why, in the random oracle model, it is possible to obtain a non-interactive commitment scheme.

Answer (i). Let $\mathbf{Adv}^{\mathsf{RO}}$ be an adversary that makes at most q queries.

WLOG, assume that $\mathbf{Adv}^{\mathsf{RO}}$ always makes exactly q RO queries. This assumption does not lead to any loss of generality because we can otherwise construct another adversary $\mathbf{Adv}^{\mathsf{RO}}_{\text{q-query}}$ functioning the same as $\mathbf{Adv}^{\mathsf{RO}}$, except that whenever $\mathbf{Adv}^{\mathsf{RO}}$ outputs an answer after making $q' < q$ RP queries, it makes $q - q'$ more "dummy" RO queries to meet our requirement, which allows $\mathbf{Adv}^{\mathsf{RO}}_{\text{q-query}}$ to achieve an equal win rate as $\mathbf{Adv}^{\mathsf{RO}}$ in the hiding security game, and then we may replace $\mathbf{Adv}^{\mathsf{RO}}$ with $\mathbf{Adv}^{\mathsf{RO}}_{\text{q-query}}$.

Moreover, we assume WLOG that all RO queries made by $\mathbf{Adv}^{\mathsf{RO}}$ are distinct. This is also WLOG because we can otherwise construct another adversary $\mathbf{Adv}^{\mathsf{RO}}_{\text{distinct}}$ functioning the same as $\mathbf{Adv}^{\mathsf{RO}}$, except that $\mathbf{Adv}^{\mathsf{RO}}_{\text{distinct}}$ records by itself all the past queries, along with the corresponding answers, and answer to itself on every query that has been answered previously, which still gives $\mathbf{Adv}^{\mathsf{RO}}_2$ an equal win rate as $\mathbf{Adv}^{\mathsf{RO}}$ in the hiding security game, and then we may replace $\mathbf{Adv}^{\mathsf{RO}}$ with $\mathbf{Adv}^{\mathsf{RO}}_{\text{distinct}}$.

Define $x_1, x_2, \ldots, x_q \in \{0,1\}^{n+1}$ as the distinct RO queries (which are random variables of course) made by $\mathbf{Adv}^{\mathsf{RO}}$ in order. Consider the event on which $x_i \neq b \parallel r$, where $b \xleftarrow{\$} \{0,1\}$ and

$r \xleftarrow{\$} \{0,1\}^n$ are sampled by the challenger and $\mathsf{com} = \mathsf{RO}(b \,\|\, r)$ is sent as input to $\mathbf{Adv}^{\mathsf{RO}}$ at the start of the game.

Recall that for any $x \in \{0,1\}^{n+1}$, the value of $\mathsf{RO}(x)$ is fresh and uniform to $\mathbf{Adv}^{\mathsf{RO}}$ if $\mathbf{Adv}^{\mathsf{RO}}$ has never obtained $\mathsf{RO}(x)$ in any way, either by querying x to $\mathsf{RO}(\cdot)$ or by receiving $\mathsf{RO}(x)$ as input. Therefore, for each query x_i, as long as $x_1, \ldots, x_i$ (current and all past queries) are distinct (which is known) and neither of them is $b \,\|\, r$, then the query answer $\mathsf{RO}(x_i)$ is fresh (to $\mathbf{Adv}^{\mathsf{RO}}$), so $\mathsf{RO}(x_i)$ must be independent of $b \,\|\, r$. Likewise, before $\mathbf{Adv}^{\mathsf{RO}}$ ever queries $b \,\|\, r$ to $\mathsf{RO}(\cdot)$, $\mathsf{com} = \mathsf{RO}(b \,\|\, r)$ always looks fresh to $\mathbf{Adv}^{\mathsf{RO}}$, so the input com must be independent of $b \,\|\, r$. Combing the two observations, it's not hard to see that before the $i+1$th query by $\mathbf{Adv}^{\mathsf{RO}}$, given $x_j \neq b \,\|\, r \;\; \forall j \in [i]$, the input com and all previous query answers $\mathsf{RO}(x_1), \ldots, \mathsf{RO}(x_i)$ are independent of $b \,\|\, r$. That is to say, for all $i \in \{0, 1, \ldots, q\}$,

$$x_1 \neq b \,\|\, r \wedge \cdots \wedge x_i \neq b \,\|\, r \implies (\mathsf{com}, \mathsf{RO}(x_1), \ldots, \mathsf{RO}(x_i)) \perp\!\!\!\perp b \,\|\, r. \tag{3.1}$$

Lemma 3.1. Let $X_1, \ldots, X_n$ be random variables. If $X_1, \ldots, X_n$ are jointly independent, then so are $f_1(X_1), \ldots, f_n(X_n)$ for any randomized functions $f_1, \ldots, f_n$, where the internal randomness of each f_i is independent of all other random variables.

Here is the interesting part: for each $i \in [q]$, by Lemma 3.1, if the input com and the first i query answers $\mathsf{RO}(x_1), \ldots, \mathsf{RO}(x_{i-1})$ are independent of $b \,\|\, r$, then the ith query x_i must be independent of $b \,\|\, r$ as well. This is because x_i can be written as a function of input and previous answers:

$$x_i = \mathbf{Adv}_i^{\mathsf{RO}}(\mathsf{com}, \mathsf{RO}(x_1), \ldots, \mathsf{RO}(x_{i-1}); r),$$

where r is the internal randomness of $\mathbf{Adv}^{\mathsf{RO}}$. Similarly, if the input com and all query answers $\mathsf{RO}(x_1), \ldots, \mathsf{RO}(x_q)$ are independent of $b \,\|\, r$, then the final output b' of $\mathbf{Adv}^{\mathsf{RO}}$ must be independent of $b \,\|\, r$ as well. Formally speaking, for $i \in [q]$,

$$(\mathsf{com}, \mathsf{RO}(x_1), \ldots, \mathsf{RO}(x_{i-1})) \perp\!\!\!\perp b \,\|\, r \implies x_i \perp\!\!\!\perp b \,\|\, r \tag{3.2}$$

$$(\mathsf{com}, \mathsf{RO}(x_1), \ldots, \mathsf{RO}(x_q)) \perp\!\!\!\perp b \,\|\, r \implies b' \perp\!\!\!\perp b \,\|\, r. \tag{3.3}$$

Combining (3.1) and (3.2) (resp. (3.3)), we deduce that for all $i \in [q]$,

$$x_1 \neq b \,\|\, r \wedge \cdots \wedge x_{i-1} \neq b \,\|\, r \implies x_i \perp\!\!\!\perp b \,\|\, r \tag{3.4}$$

$$x_1 \neq b \,\|\, r \wedge \cdots \wedge x_q \neq b \,\|\, r \implies b' \perp\!\!\!\perp b \,\|\, r. \tag{3.5}$$

We can thus bound below the probability that $\mathbf{Adv}^{\mathsf{RO}}$ has never queried $b \,\|\, r$ to $\mathsf{RO}(\cdot)$:

$$\begin{aligned}
&\Pr\left[x_1 \neq b \,\|\, r \wedge \cdots \wedge x_q \neq b \,\|\, r\right] \\
&= \Pr\left[x_1 \neq b \,\|\, r\right] \Pr\left[x_2 \neq b \,\|\, r \mid x_1 \neq b \,\|\, r\right] \cdots \Pr\left[x_q \neq b \,\|\, r \mid x_1 \neq b \,\|\, r \wedge \cdots \wedge x_{q-1} \neq b \,\|\, r\right] \\
&= \prod_{i=1}^{q} \Pr_{b \xleftarrow{\$} \{0,1\},\, r \xleftarrow{\$} \{0,1\}^n} \left[x_i \neq b \,\|\, r \mid x_1 \neq b \,\|\, r \wedge \cdots \wedge x_{i-1} \neq b \,\|\, r\right] \\
&= \prod_{i=1}^{q} \left(1 - \frac{1}{2^{n+1}}\right) \qquad (\because x_1 \neq b \,\|\, r \wedge \cdots \wedge x_{i-1} \neq b \,\|\, r \text{ implies } x_i \perp\!\!\!\perp b \,\|\, r \text{ by (3.4), while } b \,\|\, r \in \{0,1\}^{n+1} \text{ is uniform}) \\
&= \left(1 - \frac{1}{2^{n+1}}\right)^q \geq 1 - \frac{q}{2^{n+1}}. \qquad (\text{by } \textbf{Bernoulli's inequality}\text{: } (1-x)^r \geq 1 - rx \;\; \forall r \geq 1,\; x \in [0,1].)
\end{aligned} \tag{3.6}$$

Lastly, we reason

$$\begin{aligned}
\Pr[\mathbf{Adv}^{\mathsf{RO}} \text{ wins}] &= \Pr[b' = b] \\
&= \Pr\left[x_1 \neq b \,\|\, r \wedge \cdots \wedge x_q \neq b \,\|\, r\right] \underbrace{\Pr\left[b' = b \mid x_1 \neq b \,\|\, r \wedge \cdots \wedge x_q \neq b \,\|\, r\right]}_{= \frac{1}{2} \text{ (since } b' \perp\!\!\!\perp b \| r \text{ by (3.3) and } b \text{ is uniform)}} \\
&\quad + \left(1 - \Pr\left[x_1 \neq b \,\|\, r \wedge \cdots \wedge x_q \neq b \,\|\, r\right]\right) \underbrace{\Pr\left[b' = b \mid \neg(x_1 \neq b \,\|\, r \wedge \cdots \wedge x_q \neq b \,\|\, r)\right]}_{\leq 1} \\
&\leq 1 - \frac{1}{2} \Pr\left[x_1 \neq b \,\|\, r \wedge \cdots \wedge x_q \neq b \,\|\, r\right] \\
&\leq 1 - \frac{1}{2}\left(1 - \frac{q}{2^{n+1}}\right) \qquad \text{(by (3.6))} \\
&= \frac{1}{2} + \frac{q}{2^{n+2}} \\
&\leq \frac{1}{2} + \frac{2q}{2^n}.
\end{aligned}$$

⊛

Answer (ii). Let $\mathbf{Adv}^{\mathsf{RO}}$ be an adversary that makes at most q queries. Following the same reasoning as in (i), we assume WLOG that (1) $\mathbf{Adv}^{\mathsf{RO}}$ always makes exactly q RO queries and that (2) all RO queries made by $\mathbf{Adv}^{\mathsf{RO}}$ are distinct. Define $x_1, x_2, \ldots, x_q \in \{0,1\}^{n+1}$ as the distinct RO queries (which are random variables of course) made by $\mathbf{Adv}^{\mathsf{RO}}$ in order.

In fact, to win the binding game, $\mathbf{Adv}^{\mathsf{RO}}$ only has to find $r_0, r_1 \in \{0,1\}^n$ such that $\mathsf{RO}(0 \,\|\, r_0) = \mathsf{RO}(1 \,\|\, r_1)$, as $\mathbf{Adv}^{\mathsf{RO}}$ can set $\mathsf{com} := \mathsf{RO}(0 \,\|\, r_0)$ to satisfy the first equation ($\mathsf{com} = \mathsf{Com}^{\mathsf{RO}}(0; r_0)$). Simply put,

$$\begin{aligned}
&\Pr\left[\mathsf{com} = \mathsf{Com}^{\mathsf{RO}}(0; r_0) = \mathsf{Com}^{\mathsf{RO}}(1, r_1) : (\mathsf{com}, r_0, r_1) \leftarrow \mathbf{Adv}^{\mathsf{RO}}()\right] \\
&= \Pr\left[\mathsf{com} = \mathsf{RO}(0 \,\|\, r_0) = \mathsf{RO}(1 \,\|\, r_1) : (\mathsf{com}, r_0, r_1) \leftarrow \mathbf{Adv}^{\mathsf{RO}}()\right] \\
&\leq \Pr\left[\mathsf{RO}(0 \,\|\, r_0) = \mathsf{RO}(1 \,\|\, r_1) : (\cdot, r_0, r_1) \leftarrow \mathbf{Adv}^{\mathsf{RO}}()\right],
\end{aligned}$$

so it suffices to show that

$$\Pr\left[\mathsf{RO}(0 \,\|\, r_0) = \mathsf{RO}(1 \,\|\, r_1) : (\cdot, r_0, r_1) \leftarrow \mathbf{Adv}^{\mathsf{RO}}()\right] \leq \frac{1}{2^{3n}} + \frac{q^2}{2^{3n}}. \tag{3.7}$$

There are 2 cases:

- **Case 1**: $\mathbf{Adv}^{\mathsf{RO}}$ has not queried both $0 \| r_0$ and $1 \| r_1$ to $\mathsf{RO}(\cdot)$, i.e., $\{0 \| r_0, 1 \| r_1\} \not\subseteq \{x_1, \ldots, x_q\}$. WLOG, assume that $0 \,\|\, r_0$ has not been queried by $\mathbf{Adv}^{\mathsf{RO}}$ throughout its execution. This means that, in the sense of lazy sampling, $y \triangleq \mathsf{RO}(0 \,\|\, r_0) \xleftarrow{\$} \{0,1\}^{3n}$ is fresh and uniform when $\mathsf{RO}(0 \,\|\, r_0) \stackrel{?}{=} \mathsf{RO}(1 \,\|\, r_1)$ is checked. (*fresh* means being independent of all the other random variables.) Therefore, the probability that $\mathsf{RO}(1 \,\|\, r_1)$ happens to be $y \xleftarrow{\$} \{0,1\}^{3n}$ is only $\frac{1}{2^{3n}}$, i.e.,

$$\Pr\left[\mathsf{RO}(0 \,\|\, r_0) = \mathsf{RO}(1 \,\|\, r_1) \mid 0 \,\|\, r_0 \text{ is not queried by } \mathbf{Adv}^{\mathsf{RO}}\right] = \Pr_{y \xleftarrow{\$} \{0,1\}^{3n}}\left[y = \mathsf{RO}(1 \,\|\, r_1)\right] = \frac{1}{2^{3n}} \leq \frac{1}{2^{3n}} + \frac{q^2}{2^{3n}}.$$

- **Case 2:** $\mathbf{Adv}^{\mathsf{RO}}$ has queried both $0 \| r_0$ and $1 \| r_1$ to $\mathsf{RO}(\cdot)$, i.e., $\{0 \| r_0, 1 \| r_1\} \subseteq \{x_1, \ldots, x_q\}$. In the sense of lazy sampling, since every RO query x_i is distinct, every query answer $\mathsf{RO}(x_i)$ is fresh and uniformly sampled from $\{0,1\}^{3n}$, meaning that $\mathsf{RO}(x_1), \ldots, \mathsf{RO}(x_q) \overset{\text{iid}}{\sim} U_{3n}$. Also, fixing the values of $0 \| r_0$ and $1 \| r_1$, since $0 \| r_0 \neq 1 \| r_1$, they must be different queries among $x_1, \ldots, x_q$. (This means that $0 \| r_0 = x_{i_0}$ and $0 \| r_0 = x_{i_1}$ for random variables i_0, i_1 over $[q]$ where $\Pr[i_0 \neq i_1] = 1$.) We reason that if $\mathsf{RO}(0 \| r_0) = \mathsf{RO}(1 \| r_1)$, then, as $0 \| r_0$ and $1 \| r_1$ are equal to different queries among $x_1, \ldots, x_q$, there must be 2 distinct queries x_i, x_j (where $i \neq j$) that give the same RO answer, i.e.,

$$\exists i \neq j \in [q].\ \mathsf{RO}(x_i) = \mathsf{RO}(x_j).$$

This is because otherwise, fixing the values of $0 \| r_0$ and $1 \| r_1$, we would have $\mathsf{RO}(0 \| r_0) \neq \mathsf{RO}(1 \| r_1)$ as $0 \| r_0$ and $1 \| r_1$ are distinct and are both queried by $\mathbf{Adv}^{\mathsf{RO}}$. Left to be done are some dirty calculations:

$$\begin{aligned}
&\Pr\left[\mathsf{RO}(0 \| r_0) = \mathsf{RO}(1 \| r_1) \,\middle|\, 0 \| r_0 \text{ and } 1 \| r_1 \text{ are both queried by } \mathbf{Adv}^{\mathsf{RO}}\right] \\
\leq &\Pr\left[\mathsf{RO}(x_{i_0}) = \mathsf{RO}(x_{i_1}) \text{ for some r.v. } i_0, i_1 \in [q] \text{ s.t. } i_0 \neq i_1\right] \\
\leq &\Pr\left[\exists i \neq j \in [q].\mathsf{RO}(x_i) = \mathsf{RO}(x_j)\right] \\
= &\Pr\left[\bigvee_{i<j} \mathsf{RO}(x_i) = \mathsf{RO}(x_j)\right] \\
\leq &\sum_{i<j} \Pr\left[\mathsf{RO}(x_i) = \mathsf{RO}(x_j)\right] \\
= &\sum_{i<j} \frac{1}{2^{3n}} \qquad\qquad (\because i \neq j \wedge \mathsf{RO}(x_1), \ldots, \mathsf{RO}(x_q) \overset{\text{iid}}{\sim} U_{3n}) \\
= &\frac{q(q-1)}{2} \cdot \frac{1}{2^{3n}} \\
\leq &\frac{q^2}{2^{3n+1}} \leq \frac{q^2}{2^{3n}} \leq \frac{1}{2^{3n}} + \frac{q^2}{2^{3n}}.
\end{aligned}$$

In either case, we have $\Pr\left[\mathsf{RO}(0 \| r_0) = \mathsf{RO}(1 \| r_1) \mid \cdot\right] \leq \frac{1}{2^{3n}} + \frac{q^2}{2^{3n}}$. Combining the two cases we can clearly see that (3.7) holds.

> Remark. Since the domain of RO has size 2^{n+1} and we've assumed all queries to be distinct, it's natural to have $q \leq 2^{n+1}$. That is, there's nothing left to query after $\mathbf{Adv}^{\mathsf{RO}}$ has queried all 2^{n+1} inputs to RO in $\{0,1\}^{n+1}$. In this sense, we have an upper bound in absence of q:
>
> $$\Pr\left[\mathsf{RO}(0 \| r_0) = \mathsf{RO}(1 \| r_1) : (\cdot, r_0, r_1) \leftarrow \mathbf{Adv}^{\mathsf{RO}}()\right] \leq \frac{1}{2^{3n}} + \frac{q^2}{2^{3n}} \leq \frac{1}{2^{3n}} + \frac{1}{2^{n-2}}.$$

⊛

Answer (iii).

For convenience, denote as scheme A Naor's bit commitment scheme introduced in HW2, which uses PRG to achieve hiding and binding properties, and as scheme B the bit commitment scheme using RO introduced in this problem.

(a) • Syntax: Scheme A is interactive, whereas scheme B is non-interactive. The reason scheme A has to be interactive is that the scheme relies on the receiver-selected $r \in \{0,1\}^{3n}$ that is out of the control of the committer so that it is extremely hard for any adversarial committer to find a commitment (along with two decommitments) that can be opened to both bits in the reveal phase; in other words, the commitment (output of PRG padded by r) is almost always bound to a unique committed bit. By contrast, scheme B does not need padding because the essential property of RO–values being deterministic on queried values yet uniformly random on unqueried values–already achieves (computational) hiding and binding properties as we “hide” the committed bit in the input to RO. Note that solely having PRG, we cannot “hide” the committed bit in PRG’s input because PRG is not guaranteed to hide its input (e.g., it may expose partial input in its output).

- Security: Scheme A and scheme B are both computational hiding and statistically binding.
- Hiding property of interactive commitment schemes (e.g. scheme A) is defined by the absence of a (unbounded/PPT) distinguisher that distinguishes between views of the receiver when the committer committing different bits (i.e. when $b = 0$ and $b = 1$) with nonnegligible advantage. On the other hand, hiding property of non-interactive commitment schemes (e.g. scheme B) is simply defined by hardness of the corresponding hiding game, i.e., by the absence of a (unbounded/PPT) adversarial receiver that infers the committed bit by just looking at the commitment with nonnegligible advantage.

(b)
- Hiding:
  - Scheme A achieves hiding property by exploiting the security given by PRG–as long as the input is uniformly random, the output, which is used to pad 0^{3n} when committing $b = 0$ and r (selected by the receiver) when committing $b = 1$, must be pseudorandom, i.e., computationally indistinguishable to a uniformly random padding. Since the receiver does not know the input to PRG that derives the pseudorandom padding, it never knows whether it was 0^{3n} (meaning $b = 0$) or r (meaning $b = 1$) that got padded by just looking at the padded result.
  - Scheme B achieves hiding property simply by hiding the committed bit, with some randomness appended, in the input of RO. Since it is highly unlikely for any PPT adversarial receiver to query to RO the same bit with the exact randomness, any PPT adversarial receiver can guess the committed bit with only negligible advantage.
- Binding:
  - Scheme A achieves binding property by using a PRG whose codomain is exponentially larger than the square of the size of its domain. We’ve already shown why this requirement results in statistical binding in HW2. Simply put, this is because the fact that the receiver-selected $r \in \{0, 1\}^{3n}$ renders it nearly impossible (i.e., happening with negligible probability) that a commitment opening to both bits exists.
  - Scheme B achieves binding property by exploiting the property of random oracle: every output is uniformly random if it has not been queried before. This nearly exterminates any commitment that, when provided with a pair of randomness, opens to both bits., i.e., such “bad” commitment exists with merely negligible probability.

(c) The use of RO enables the construction of a non-interactive commitment scheme because it allows us to replace the receiver-selected parameters (which are out of the control of the committer and thus used to achieve binding property) with the internal randomness of RO, whose outputs are selected purely out of randomness (or rather, by God) and are thus out of the control of any (potentially-malicious) committer. Still, once queried, outputs of RO are then fixed, allowing the input to RO to be served as the opening information (aka decommitment). ⊛

Problem 4 (From KEM to PKE).

In class, we've shown how to achieve an IND-CCA secure KEM via FO transformation. In this problem, we show that we can upgrade it to an IND-CCA secure PKE with the help of an IND-CCA secure SKE scheme.

We show how to obtain an IND-CCA secure PKE scheme PKE via:

- An IND-CCA secure KEM scheme $\mathsf{KEM} = (\mathsf{KeyGen}_{\mathsf{KEM}}, \mathsf{Encaps}, \mathsf{Decaps})$, where the output key $k \in \{0,1\}^{\ell(n)}$ for security parameter 1^n.
- An IND-CCA secure SKE scheme $\mathsf{SKE} = (\mathsf{KeyGen}_{\mathsf{SKE}}, \mathsf{Enc}_{\mathsf{SKE}}, \mathsf{Dec}_{\mathsf{SKE}})$, where $\mathsf{KeyGen}_{\mathsf{SKE}}(1^n)$ uniformly samples $k \leftarrow \{0,1\}^{\ell(n)}$. The message space on security parameter 1^n is defined as $\mathcal{M}_n$.

The combined PKE scheme $\mathsf{PKE} = (\mathsf{KeyGen}_{\mathsf{PKE}}, \mathsf{Enc}_{\mathsf{PKE}}, \mathsf{Dec}_{\mathsf{PKE}})$ with message space also $\mathcal{M}_n$ is defined as follows:

$\mathsf{KeyGen}_{\mathsf{PKE}}(1^n)$	$\mathsf{Enc}_{\mathsf{PKE}}(\mathsf{pk}, m \in \mathcal{M}_n)$	$\mathsf{Dec}_{\mathsf{PKE}}(\mathsf{sk}, \mathsf{ct}^*_{\mathsf{PKE}} = (\mathsf{ct}^*_{\mathsf{KEM}}, \mathsf{ct}^*_{\mathsf{SKE}}))$
1: **return** $\mathsf{KeyGen}_{\mathsf{KEM}}(1^n)$	1: $(\mathsf{ct}_{\mathsf{KEM}}, k) \leftarrow \mathsf{Encaps}(\mathsf{pk})$	1: $k^* := \mathsf{Decaps}_{\mathsf{sk}}(\mathsf{ct}^*_{\mathsf{KEM}})$
	2: $\mathsf{ct}_{\mathsf{SKE}} \leftarrow \mathsf{Enc}_{\mathsf{SKE}}(k, m)$	2: **if** $k^* = \perp$ **then**
	3: **return** $(\mathsf{ct}_{\mathsf{KEM}}, \mathsf{ct}_{\mathsf{SKE}})$	3: **return** $\perp$
		4: **else**
		5: **return** $\mathsf{Dec}_{\mathsf{SKE}}(k^*, \mathsf{ct}^*_{\mathsf{SKE}})$

Show that the previous construction is indeed an IND-CCA secure PKE scheme.
(**Hint.** In order to play with the PKE adversary, you will need to simulate the decryption oracle. It is natural to see that you will need the decryption oracle from the SKE scheme. How can you bridge the two interface? To achieve this, you will have to somehow make use of the IND-CCA property of the KEM scheme.)

Answer. We prove correctness first, followed by a proof of IND-CCA security of the scheme PKE.

Proof of Correctness.

Since both KEM and SKE are correct (w.r.t. their own correctness definitions), we know that there exist negligible functions $\mu(\cdot), \varepsilon(\cdot)$ such that for every $n \in \mathbb{N}$ and every message $m \in \mathcal{M}_n$,

$$\begin{cases} \Pr_{(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{KeyGen}_{\mathsf{KEM}}(1^n)} [(\mathsf{ct}, k) \leftarrow \mathsf{Encaps}(\mathsf{pk}) : \mathsf{Decaps}_{\mathsf{sk}}(\mathsf{ct}) = k] & \geq 1 - \mu(n) \\ \Pr_{k \leftarrow \mathsf{KeyGen}_{\mathsf{SKE}}(1^n)} [\mathsf{Dec}_{\mathsf{SKE}}(k, \mathsf{Enc}_{\mathsf{SKE}}(k, m)) = m] = \Pr_{k \xleftarrow{\$} \{0,1\}^{\ell(n)}} [\mathsf{Dec}_{\mathsf{SKE}}(k, \mathsf{Enc}_{\mathsf{SKE}}(k, m)) = m] & \geq 1 - \varepsilon(n) \end{cases} \tag{4.1}$$

Therefore, for every $n \in \mathbb{N}$ and every message $m \in \mathcal{M}_n$, we have

$$
\begin{aligned}
&\Pr_{(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{KeyGen}_{\mathsf{PKE}}(1^n)} [\mathsf{Dec}_{\mathsf{PKE}}(\mathsf{pk}, \mathsf{Enc}_{\mathsf{PKE}}(\mathsf{pk}, m)) = m] \\
=& \Pr_{(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{KeyGen}_{\mathsf{KEM}}(1^n)} \left[\begin{array}{r} (\mathsf{ct}_{\mathsf{KEM}}, k) \leftarrow \mathsf{Encaps}(\mathsf{pk}) \\ k^* \triangleq \mathsf{Decaps}_{\mathsf{sk}}(\mathsf{ct}_{\mathsf{KEM}}) \end{array} : \begin{Bmatrix} \perp & \text{if } k^* = \perp \\ \mathsf{Dec}_{\mathsf{SKE}}(k^*, \mathsf{Enc}_{\mathsf{SKE}}(k, m)) & \text{otherwise} \end{Bmatrix} = m \right] \\
\geq& \Pr_{(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{KeyGen}_{\mathsf{KEM}}(1^n)} [(\mathsf{ct}_{\mathsf{KEM}}, k) \leftarrow \mathsf{Encaps}(\mathsf{pk}) : \mathsf{Decaps}_{\mathsf{sk}}(\mathsf{ct}_{\mathsf{KEM}}) = k \wedge \mathsf{Dec}_{\mathsf{SKE}}(k, \mathsf{Enc}_{\mathsf{SKE}}(k, m)) = m] \quad \text{(by implication)} \\
\geq& 1 - \left(1 - \Pr_{(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{KeyGen}_{\mathsf{KEM}}(1^n)} [(\mathsf{ct}_{\mathsf{KEM}}, k) \leftarrow \mathsf{Encaps}(\mathsf{pk}) : \mathsf{Decaps}_{\mathsf{sk}}(\mathsf{ct}_{\mathsf{KEM}}) = k]\right) \\
&\quad - \Bigg(1 - \underbrace{\Pr_{(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{KeyGen}_{\mathsf{KEM}}(1^n)} [(\cdot, k) \leftarrow \mathsf{Encaps}(\mathsf{pk}) : \mathsf{Dec}_{\mathsf{SKE}}(k, \mathsf{Enc}_{\mathsf{SKE}}(k, m)) = m]}_{\geq \Pr_{k \xleftarrow{\$} \{0,1\}^{\ell(n)}} [\mathsf{Dec}_{\mathsf{SKE}}(k, \mathsf{Enc}_{\mathsf{SKE}}(k, m)) = m] - \mathsf{negl}(n) \ \text{(by Claim 4.1)}}\Bigg) \quad \text{(by Union Bound)} \\
\geq& 1 - (1 - (1 - \mu(n))) - (1 - ((1 - \varepsilon(n)) - \mathsf{negl}(n))) \qquad \text{(by (4.1))} \\
=& 1 - \mu(n) - \varepsilon(n) - \mathsf{negl}(n) \\
=& 1 - \mathsf{negl}(n) - \mathsf{negl}(n) - \mathsf{negl}(n) = 1 - \mathsf{negl}(n),
\end{aligned}
$$

which implies that PKE $= (\mathsf{KeyGen}_{\mathsf{PKE}}, \mathsf{Enc}_{\mathsf{PKE}}, \mathsf{Dec}_{\mathsf{PKE}})$ is a correct PKE scheme.

It suffices to show Claim 4.1 indeed holds, which implies that for all $n \in \mathbb{N}$ and every message $m \in \mathcal{M}_n$,

$$
\Pr_{\substack{(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{KeyGen}_{\mathsf{KEM}}(1^n) \\ (\cdot, k) \leftarrow \mathsf{Encaps}(\mathsf{pk})}} [\mathsf{Dec}_{\mathsf{SKE}}(k, \mathsf{Enc}_{\mathsf{SKE}}(k, m)) = m] \geq \Pr_{k \xleftarrow{\$} \{0,1\}^{\ell(n)}} [\mathsf{Dec}_{\mathsf{SKE}}(k, \mathsf{Enc}_{\mathsf{SKE}}(k, m)) = m] - \mathsf{negl}(n),
$$

as used in the reasoning above.

Claim 4.1. There exists a negligible function $\nu(\cdot)$ such that for all $n \in \mathbb{N}$ and every message $m \in \mathcal{M}_n$,

$$
\left| \Pr_{\substack{(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{KeyGen}_{\mathsf{KEM}}(1^n) \\ (\cdot, k) \leftarrow \mathsf{Encaps}(\mathsf{pk})}} [\mathsf{Dec}_{\mathsf{SKE}}(k, \mathsf{Enc}_{\mathsf{SKE}}(k, m)) = m] - \Pr_{k \xleftarrow{\$} \{0,1\}^{\ell(n)}} [\mathsf{Dec}_{\mathsf{SKE}}(k, \mathsf{Enc}_{\mathsf{SKE}}(k, m)) = m] \right| \leq \nu(n).
$$

Proof. For convenience, denote

$$
\Delta_n \triangleq \Pr_{\substack{(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{KeyGen}_{\mathsf{KEM}}(1^n) \\ (\cdot, k) \leftarrow \mathsf{Encaps}(\mathsf{pk})}} [\mathsf{Dec}_{\mathsf{SKE}}(k, \mathsf{Enc}_{\mathsf{SKE}}(k, m)) = m] - \Pr_{k \xleftarrow{\$} \{0,1\}^{\ell(n)}} [\mathsf{Dec}_{\mathsf{SKE}}(k, \mathsf{Enc}_{\mathsf{SKE}}(k, m)) = m] .
$$

Suppose to the contrary that the claim does not hold, meaning that there exist $c > 0$ and an ensemble $\{m_n^*\}_{n \in \mathbb{N}}$ of messages, where $m_n^* \in \mathcal{M}_n$ for each $n \in \mathbb{N}$, such that

$$
|\Delta_n| \geq \frac{1}{n^c} \quad \text{for infinitely many } n \in \mathbb{N} \text{ (or more precisely, for all } n \in S \text{ where } S \subseteq \mathbb{N} \text{ is infinite).} \tag{4.2}
$$

In fact, it'd be better to know on which $n \in \mathbb{N}$ we have $|\Delta_n| \geq \frac{1}{n^c}$, so we explicitly define (possibly infinite) subsets of $\mathbb{N}$:

$$
S \triangleq \left\{ n \in \mathbb{N} \;\middle|\; |\Delta_n| \geq \frac{1}{n^c} \right\}, \quad S_+ \triangleq \{n \in S \mid \Delta_n > 0\}, \quad S_- \triangleq \{n \in S \mid \Delta_n < 0\} .
$$

Since $S = S_+ \cup S_-$ and S is infinite by (4.2), either S_+ or S_- is infinite. ■

cont'd. We can then construct PPT oracle adversaries (a reduction) $\mathbf{Adv}, \mathbf{Adv}'$ such that either of the two breaks the IND-CCA security of $\mathsf{KEM} = (\mathsf{KeyGen}_{\mathsf{KEM}}, \mathsf{Encaps}, \mathsf{Decaps})$ as follows:

$\mathbf{Adv}^{\mathsf{Decaps}_{\mathsf{sk}}(\cdot)}(\mathsf{pk}, \mathsf{ct}, k_b)$	$\mathbf{Adv}'^{\mathsf{Decaps}_{\mathsf{sk}}(\cdot)}(\mathsf{pk}, \mathsf{ct}, k_b)$
1: **if** $\mathsf{Dec}_{\mathsf{SKE}}(k_b, \mathsf{Enc}_{\mathsf{SKE}}(k_b, m_n^*)) = m_n^*$ **then**	1: **return** $1 - \mathbf{Adv}^{\mathsf{Decaps}_{\mathsf{sk}}(\cdot)}(\mathsf{pk}, \mathsf{ct}, k_b)$
2: **return** $b' := 0$	
3: **else**	
4: **return** $b' := 1$	

Clearly $\mathbf{Adv}$ runs in PPT. For all $n \in S_+ \subseteq \mathbb{N}$, in the IND-CCA experiment $\mathsf{KEM}^{\mathsf{IND-CCA}}$ of KEM, the probability that $\mathbf{Adv}$ wins is:

$$
\begin{aligned}
&\Pr[\mathbf{Adv} \text{ wins } \mathsf{KEM}^{\mathsf{IND-CCA}}] \\
=&\Pr_{\substack{(\mathsf{pk},\mathsf{sk})\leftarrow\mathsf{KeyGen}_{\mathsf{KEM}}(1^n)\\ (\mathsf{ct},k_0)\leftarrow\mathsf{Encaps}(\mathsf{pk})\\ k_1\xleftarrow{\$}\{0,1\}^{\ell(n)}\\ b\xleftarrow{\$}\{0,1\}}}\left[\mathbf{Adv}^{\mathsf{Decaps}_{\mathsf{sk}}(\cdot)}(\mathsf{pk},\mathsf{ct},k_b)=b\right] \\
=&\frac{1}{2}\Pr_{\substack{(\mathsf{pk},\mathsf{sk})\leftarrow\mathsf{KeyGen}_{\mathsf{KEM}}(1^n)\\ (\mathsf{ct},k_0)\leftarrow\mathsf{Encaps}(\mathsf{pk})}}\left[\mathbf{Adv}^{\mathsf{Decaps}_{\mathsf{sk}}(\cdot)}(\mathsf{pk},\mathsf{ct},k_0)=0\right]+\frac{1}{2}\Pr_{\substack{(\mathsf{pk},\mathsf{sk})\leftarrow\mathsf{KeyGen}_{\mathsf{KEM}}(1^n)\\ (\mathsf{ct},k_0)\leftarrow\mathsf{Encaps}(\mathsf{pk})\\ k_1\xleftarrow{\$}\{0,1\}^{\ell(n)}}}\left[\mathbf{Adv}^{\mathsf{Decaps}_{\mathsf{sk}}(\cdot)}(\mathsf{pk},\mathsf{ct},k_1)=1\right] \\
=&\frac{1}{2}\Pr_{\substack{(\mathsf{pk},\mathsf{sk})\leftarrow\mathsf{KeyGen}_{\mathsf{KEM}}(1^n)\\ (\cdot,k_0)\leftarrow\mathsf{Encaps}(\mathsf{pk})}}[\mathsf{Dec}_{\mathsf{SKE}}(k_0,\mathsf{Enc}_{\mathsf{SKE}}(k_0,m_n^*))=m_n^*]+\frac{1}{2}\Pr_{k_1\xleftarrow{\$}\{0,1\}^{\ell(n)}}[\mathsf{Dec}_{\mathsf{SKE}}(k_1,\mathsf{Enc}_{\mathsf{SKE}}(k_1,m_n^*))\neq m_n^*] \\
=&\frac{1}{2}+\frac{1}{2}\left(\Pr_{\substack{(\mathsf{pk},\mathsf{sk})\leftarrow\mathsf{KeyGen}_{\mathsf{KEM}}(1^n)\\ (\cdot,k_0)\leftarrow\mathsf{Encaps}(\mathsf{pk})}}[\mathsf{Dec}_{\mathsf{SKE}}(k_0,\mathsf{Enc}_{\mathsf{SKE}}(k_0,m_n^*))=m_n^*]-\Pr_{k_1\xleftarrow{\$}\{0,1\}^{\ell(n)}}[\mathsf{Dec}_{\mathsf{SKE}}(k_1,\mathsf{Enc}_{\mathsf{SKE}}(k_1,m_n^*))=m_n^*]\right) \\
=&\frac{1}{2}+\frac{1}{2}\Delta_n \\
=&\frac{1}{2}+\frac{1}{2}|\Delta_n| \qquad (\because n\in S_+) \\
\geq&\frac{1}{2}+\frac{1}{2n^c}. \qquad (\text{by } (4.2))
\end{aligned}
$$

Notice that $\mathbf{Adv}'$ is basically the complement of $\mathbf{Adv}$, so for all $n \in S_- = S \setminus S_+ \subseteq \mathbb{N}$, in the IND-CCA experiment $\mathsf{KEM}^{\mathsf{IND-CCA}}$ of KEM, the probability that $\mathbf{Adv}'$ wins is:

$$
\begin{aligned}
\Pr[\mathbf{Adv}' \text{ wins } \mathsf{KEM}^{\mathsf{IND-CCA}}] = 1 - \Pr[\mathbf{Adv} \text{ wins } \mathsf{KEM}^{\mathsf{IND-CCA}}] &= 1-\left(\frac{1}{2}+\frac{1}{2}\Delta_n\right) \qquad (\text{by previous calculation}) \\
&= \frac{1}{2}-\frac{1}{2}\Delta_n \\
&= \frac{1}{2}+\frac{1}{2}|\Delta_n| \qquad (\because n\in S_-) \\
&\geq \frac{1}{2}+\frac{1}{2n^c}. \qquad (\text{by } (4.2))
\end{aligned}
$$

Recall that either S_+ or S_- is infinite. If S_+ is infinite, then from the reasoning above we know that for infinitely many $n \in \mathbb{N}$ (namely, for all $n \in S_+$), $\mathbf{Adv}$ wins $\mathsf{KEM}^{\mathsf{IND-CCA}}$ with advantage at least $\frac{1}{2n^c}$, which is nonnegligible. If S_- is infinite, then from the reasoning above we know that for infinitely many $n \in \mathbb{N}$ (namely, for all $n \in S_-$), $\mathbf{Adv}'$ wins $\mathsf{KEM}^{\mathsf{IND-CCA}}$ with advantage at least $\frac{1}{2n^c}$, which is nonnegligible. Hence, either $\mathbf{Adv}$ or $\mathbf{Adv}'$ breaks the IND-CCA security of KEM, which is a contradiction. ■

Hence the correctness of PKE is proved.

Proof of IND-CCA security.

We prove this by contradiction. Suppose to the contrary that $\mathsf{PKE} = (\mathsf{KeyGen}_{\mathsf{PKE}}, \mathsf{Enc}_{\mathsf{PKE}}, \mathsf{Dec}_{\mathsf{PKE}})$ is not IND-CCA-secure. Then there exist a PPT oracle adversary **Adv** and a nonnegligible function $\varepsilon(\cdot)$ such that in the IND-CCA experiment $\mathsf{PKE}^{\mathsf{IND-CCA}}$ of PKE (cf. Figure 8), for all $n \in \mathbb{N}$, **Adv** wins with advantage $\varepsilon(n)$, i.e.,

$$\Pr\left[\mathbf{Adv} \text{ wins } \mathsf{PKE}^{\mathsf{IND-CCA}}\right] = \frac{1}{2} + \varepsilon(n). \tag{4.3}$$

Note that, in the experiment $\mathsf{PKE}^{\mathsf{IND-CCA}}$, we have expanded the definitions of $\mathsf{PKE} = (\mathsf{KeyGen}_{\mathsf{PKE}}, \mathsf{Enc}_{\mathsf{PKE}}, \mathsf{Dec}_{\mathsf{PKE}})$ for convenience.

Ch		$\mathbf{Adv}^{\mathsf{Dec}_{\mathsf{PKE}}(\mathsf{sk},\cdot)}$
$(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}_{\mathsf{KEM}}(1^n)$		
	$\xrightarrow{\mathsf{pk}}$	
$\underline{\mathsf{Dec}_{\mathsf{PKE}}(\mathsf{sk}, \mathsf{ct}_{\mathsf{PKE}})}$ 1: Parse $(\mathsf{ct}_{\mathsf{KEM}}, \mathsf{ct}_{\mathsf{SKE}}) := \mathsf{ct}_{\mathsf{PKE}}$ 2: $k := \mathsf{Decaps}_{\mathsf{sk}}(\mathsf{ct}_{\mathsf{KEM}})$ 3: **return** $\begin{cases} \perp & \text{if } k = \perp \\ \mathsf{Dec}_{\mathsf{SKE}}(k, \mathsf{ct}_{\mathsf{SKE}}) & \text{if } k \neq \perp \end{cases}$	$\xleftarrow{\mathsf{ct}_{\mathsf{PKE}\,i}}$ $\xrightarrow{\mathsf{Dec}_{\mathsf{PKE}}(\mathsf{sk},\mathsf{ct}_{\mathsf{PKE}\,i})}$	Query $\mathsf{Dec}_{\mathsf{PKE}}(\mathsf{sk}, \cdot)$
$b \xleftarrow{\$} \{0,1\}$	$\xleftarrow{(m_0, m_1)}$	
$(\mathsf{ct}^*_{\mathsf{KEM}}, k^*) \leftarrow \mathsf{Encaps}(\mathsf{pk})$ $\mathsf{ct}^*_{\mathsf{SKE}} \leftarrow \mathsf{Enc}_{\mathsf{SKE}}(k^*, m_b)$ $\mathsf{ct}^*_{\mathsf{PKE}} := (\mathsf{ct}^*_{\mathsf{KEM}}, \mathsf{ct}^*_{\mathsf{SKE}})$	$\xrightarrow{\mathsf{ct}^*_{\mathsf{PKE}}}$	
$\underline{\mathsf{Dec}_{\mathsf{PKE}}(\mathsf{sk}, \mathsf{ct}_{\mathsf{PKE}} \neq \mathsf{ct}^*_{\mathsf{PKE}})}$ 1: Parse $(\mathsf{ct}_{\mathsf{KEM}}, \mathsf{ct}_{\mathsf{SKE}}) := \mathsf{ct}_{\mathsf{PKE}}$ 2: $k := \mathsf{Decaps}_{\mathsf{sk}}(\mathsf{ct}_{\mathsf{KEM}})$ 3: **return** $\begin{cases} \perp & \text{if } k = \perp \\ \mathsf{Dec}_{\mathsf{SKE}}(k, \mathsf{ct}_{\mathsf{SKE}}) & \text{if } k \neq \perp \end{cases}$	$\xleftarrow{\mathsf{ct}_{\mathsf{PKE}\,i} \neq \mathsf{ct}^*_{\mathsf{PKE}}}$ $\xrightarrow{\mathsf{Dec}_{\mathsf{PKE}}(\mathsf{sk},\mathsf{ct}_{\mathsf{PKE}\,i})}$	Query $\mathsf{Dec}_{\mathsf{PKE}}(\mathsf{sk}, \cdot)$
Adv wins if $b' = b$	$\xleftarrow{b'}$	

Figure 8: Experiment $\mathsf{PKE}^{\mathsf{IND-CCA}}$

Now consider a slightly different game G_1 (cf. Figure 9), where the SKE key $k^* \in \{0,1\}^{\ell(n)}$ is uniformly sampled instead of generated by Encaps. We claim that thanks to the IND-CCA security of KEM, the difference between win rates of **Adv** in experiment $\mathsf{PKE}^{\mathsf{IND-CCA}}$ and game G_1 is guaranteed to be negligible.

> Claim 4.2. There exists a negligible function $\nu(\cdot)$ such that for all $n \in \mathbb{N}$,
>
> $$\left|\Pr\left[\mathbf{Adv} \text{ wins } \mathsf{PKE}^{\mathsf{IND-CCA}}\right] - \Pr\left[\mathbf{Adv} \text{ wins } \mathsf{G}_1\right]\right| \leq \nu(n).$$

Ch		$\mathbf{Adv}^{\mathsf{Dec_{PKE}}(\mathsf{sk},\cdot)}$
$(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{KeyGen_{KEM}}(1^n)$		
$(\mathsf{ct}^*_{\mathsf{KEM}},\cdot) \leftarrow \mathsf{Encaps}(\mathsf{pk});\ k^* \xleftarrow{\$} \{0,1\}^{\ell(n)}$	$\xrightarrow{\mathsf{pk}}$	
$\boxed{\begin{array}{l} \underline{\mathsf{Dec_{PKE}}(\mathsf{sk},\mathsf{ct_{PKE}})} \\ 1\text{: Parse } (\mathsf{ct_{KEM}},\mathsf{ct_{SKE}}) \coloneqq \mathsf{ct_{PKE}} \\ 2\text{: } k \coloneqq \begin{cases} k^* & \text{if } \mathsf{ct_{KEM}} = \mathsf{ct}^*_{\mathsf{KEM}} \\ \mathsf{Decaps_{sk}}(\mathsf{ct_{KEM}}) & \text{if } \mathsf{ct_{KEM}} \neq \mathsf{ct}^*_{\mathsf{KEM}} \end{cases} \\ 3\text{: } \mathbf{return} \begin{cases} \perp & \text{if } k = \perp \\ \mathsf{Dec_{SKE}}(k,\mathsf{ct_{SKE}}) & \text{if } k \neq \perp \end{cases} \end{array}}$	$\xleftarrow{\mathsf{ct_{PKE}}_i}$ $\xrightarrow{\mathsf{Dec_{PKE}}(\mathsf{sk},\mathsf{ct_{PKE}}_i)}$	Query $\mathsf{Dec_{PKE}}(\mathsf{sk},\cdot)$
$b \xleftarrow{\$} \{0,1\}$	$\xleftarrow{(m_0,m_1)}$	
$\mathsf{ct}^*_{\mathsf{SKE}} \leftarrow \mathsf{Enc_{SKE}}(k^*,m_b)$		
$\mathsf{ct}^*_{\mathsf{PKE}} \coloneqq (\mathsf{ct}^*_{\mathsf{KEM}},\mathsf{ct}^*_{\mathsf{SKE}})$	$\xrightarrow{\mathsf{ct}^*_{\mathsf{PKE}}}$	
$\boxed{\begin{array}{l} \underline{\mathsf{Dec_{PKE}}(\mathsf{sk},\mathsf{ct_{PKE}} \neq \mathsf{ct}^*_{\mathsf{PKE}})} \\ 1\text{: Parse } (\mathsf{ct_{KEM}},\mathsf{ct_{SKE}}) \coloneqq \mathsf{ct_{PKE}} \\ 2\text{: } k \coloneqq \begin{cases} k^* & \text{if } \mathsf{ct_{KEM}} = \mathsf{ct}^*_{\mathsf{KEM}} \\ \mathsf{Decaps_{sk}}(\mathsf{ct_{KEM}}) & \text{if } \mathsf{ct_{KEM}} \neq \mathsf{ct}^*_{\mathsf{KEM}} \end{cases} \\ 3\text{: } \mathbf{return} \begin{cases} \perp & \text{if } k = \perp \\ \mathsf{Dec_{SKE}}(k,\mathsf{ct_{SKE}}) & \text{if } k \neq \perp \end{cases} \end{array}}$	$\xleftarrow{\mathsf{ct_{PKE}}_i \neq \mathsf{ct}^*_{\mathsf{PKE}}}$ $\xrightarrow{\mathsf{Dec_{PKE}}(\mathsf{sk},\mathsf{ct_{PKE}}_i)}$	Query $\mathsf{Dec_{PKE}}(\mathsf{sk},\cdot)$
$\mathbf{Adv}$ wins if $b' = b$	$\xleftarrow{b'}$	

Figure 9: Game G_1

Proof of Claim 4.2. Consider the following PPT oracle adversary R attacking the IND-CCA security of KEM in experiment $\mathsf{KEM}^{\mathsf{IND-CCA}}$:

Ch		$R^{\mathsf{Decaps_{sk}}(\cdot)}$		$\mathbf{Adv}^{\mathsf{Dec_{PKE}}(\mathsf{sk},\cdot}$
$(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{KeyGen_{KEM}}(1^n)$				
$(\mathsf{ct}^*_{\mathsf{KEM}},k_0) \leftarrow \mathsf{Encaps}(\mathsf{pk})$				
$k_1 \xleftarrow{\$} \{0,1\}^{\ell(n)}$				
$d \xleftarrow{\$} \{0,1\};\ k^* \coloneqq k_d$	$\xrightarrow{(\mathsf{pk},\mathsf{ct}^*_{\mathsf{KEM}},k^*)}$		$\xrightarrow{(\mathsf{pk},\mathsf{ct}^*_{\mathsf{KEM}},k^*)}$	
	$\xleftarrow{\mathsf{ct_{KEM}} \neq \mathsf{ct}^*_{\mathsf{KEM}}}$ $\xrightarrow{\mathsf{Decaps_{sk}}(\mathsf{ct_{KEM}})}$	$\boxed{\begin{array}{l} \underline{\mathsf{Dec_{PKE}}(\mathsf{sk},\mathsf{ct_{PKE}})} \\ 1\text{: Parse } (\mathsf{ct_{KEM}},\mathsf{ct_{SKE}}) \coloneqq \mathsf{ct_{PKE}} \\ 2\text{: } k \coloneqq \begin{cases} k^* & \text{if } \mathsf{ct_{KEM}} = \mathsf{ct}^*_{\mathsf{KEM}} \\ \mathsf{Decaps_{sk}}(\mathsf{ct_{KEM}}) & \text{if } \mathsf{ct_{KEM}} \neq \mathsf{ct}^*_{\mathsf{KEM}} \end{cases} \\ 3\text{: } \mathbf{return} \begin{cases} \perp & \text{if } k = \perp \\ \mathsf{Dec_{SKE}}(k,\mathsf{ct_{SKE}}) & \text{if } k \neq \perp \end{cases} \end{array}}$	$\xleftarrow{\mathsf{ct_{PKE}}_i}$ $\xrightarrow{\mathsf{Dec_{PKE}}(\mathsf{sk},\mathsf{ct_{PKE}}_i)}$	Query $\mathsf{Dec_{PKE}}$(sk
		$b \xleftarrow{\$} \{0,1\}$	$\xleftarrow{(m_0,m_1)}$	
		$\mathsf{ct}^*_{\mathsf{SKE}} \leftarrow \mathsf{Enc_{SKE}}(k^*,m_b)$		
		$\mathsf{ct}^*_{\mathsf{PKE}} \coloneqq (\mathsf{ct}^*_{\mathsf{KEM}},\mathsf{ct}^*_{\mathsf{SKE}})$	$\xrightarrow{\mathsf{ct}^*_{\mathsf{PKE}}}$	
	$\xleftarrow{\mathsf{ct_{KEM}} \neq \mathsf{ct}^*_{\mathsf{KEM}}}$ $\xrightarrow{\mathsf{Decaps_{sk}}(\mathsf{ct_{KEM}})}$	$\boxed{\begin{array}{l} \underline{\mathsf{Dec_{PKE}}(\mathsf{sk},\mathsf{ct_{PKE}} \neq \mathsf{ct}^*_{\mathsf{PKE}})} \\ 1\text{: Parse } (\mathsf{ct_{KEM}},\mathsf{ct_{SKE}}) \coloneqq \mathsf{ct_{PKE}} \\ 2\text{: } k \coloneqq \begin{cases} k^* & \text{if } \mathsf{ct_{KEM}} = \mathsf{ct}^*_{\mathsf{KEM}} \\ \mathsf{Decaps_{sk}}(\mathsf{ct_{KEM}}) & \text{if } \mathsf{ct_{KEM}} \neq \mathsf{ct}^*_{\mathsf{KEM}} \end{cases} \\ 3\text{: } \mathbf{return} \begin{cases} \perp & \text{if } k = \perp \\ \mathsf{Dec_{SKE}}(k,\mathsf{ct_{SKE}}) & \text{if } k \neq \perp \end{cases} \end{array}}$	$\xleftarrow{\mathsf{ct_{PKE}}_i \neq \mathsf{ct}^*_{\mathsf{PKE}}}$ $\xrightarrow{\mathsf{Dec_{PKE}}(\mathsf{sk},\mathsf{ct_{PKE}}_i)}$	Query $\mathsf{Dec_{PKE}}$(sk
R wins if $d' = d$	$\xleftarrow{d'}$	$d' \coloneqq [b' = b]$	$\xleftarrow{b'}$	

Figure 10: Experiment $\mathsf{KEM}^{\mathsf{IND-CCA}}$

■

cont'd. Clearly $\mathbf{Adv}$ runs in PPT as $\mathbf{Adv}$ also runs in PPT.

Note that in the decryption oracle $\mathsf{Dec}_{\mathsf{PKE}}(\mathsf{sk},\cdot)$ simulated by R, it is not allowed to query $\mathsf{ct}^*_{\mathsf{KEM}}$ to the decapsulation oracle $\mathsf{Decaps}_{\mathsf{sk}}(\cdot)$ provided by the challenger, so R instead answers k^* itself when $ct_{\mathsf{KEM}} = ct^*_{\mathsf{KEM}}$. We discuss the two cases of $d \in \{0,1\}$:

- If $d = 0$, then key k^* is generated by Encaps. Observe that conditioning on $\mathsf{Decaps}_{\mathsf{sk}}(\mathsf{ct}^*_{\mathsf{KEM}}) = k^*$ (i.e. the key k^* is correctly decapsulated), with the use of the abovementioned sidestep, the decryption oracle $\mathsf{Dec}_{\mathsf{PKE}}(\mathsf{sk},\cdot)$ simulated by R is in fact equivalent to the one in $\mathsf{PKE}^{\mathsf{IND-CCA}}$. Thus, the *view* of $\mathbf{Adv}$ in $\mathsf{KEM}^{\mathsf{IND-CCA}}$ is distributed identically to that of $\mathbf{Adv}$ in the original experiment $\mathsf{PKE}^{\mathsf{IND-CCA}}$.
- If $d = 1$, then key k^* is uniform and independent. Observe that the decryption oracle $\mathsf{Dec}_{\mathsf{PKE}}(\mathsf{sk},\cdot)$ simulated by R is in fact equivalent to the one in G_1. Thus, the *view* of $\mathbf{Adv}$ in $\mathsf{KEM}^{\mathsf{IND-CCA}}$ is distributed identically to that of $\mathbf{Adv}$ in the modified game G_1.

Since KEM is an IND-CCA-secure KEM and R is a PPT adversary, there exists a negligible function $\nu_1(\cdot)$ such that for all $n \in \mathbb{N}$, $\Pr\left[R \text{ wins } \mathsf{KEM}^{\mathsf{IND-CCA}}\right] \le \frac{1}{2} + \nu_1(n)$, so we have

$$
\begin{aligned}
&\frac{1}{2} + \nu_1(n) \\
\ge& \Pr\left[R \text{ wins } \mathsf{KEM}^{\mathsf{IND-CCA}}\right] \\
=& \frac{1}{2}\Pr\left[b' \ne b \mid d = 0\right] + \frac{1}{2}\Pr\left[b' = b \mid d = 1\right] \\
\ge& \frac{1}{2}\Pr\left[b' \ne b \wedge \mathsf{Decaps}_{\mathsf{sk}}(\mathsf{ct}^*_{\mathsf{KEM}}) = k^* \mid d = 0\right] + \frac{1}{2}\Pr\left[b' = b \mid d = 1\right] \\
=& \frac{1}{2}\Pr\left[\mathbf{Adv} \text{ loses and } \mathsf{Decaps}_{\mathsf{sk}}(\mathsf{ct}^*_{\mathsf{KEM}}) = k^* \text{ in } \mathsf{PKE}^{\mathsf{IND-CCA}}\right] + \frac{1}{2}\Pr\left[\mathbf{Adv} \text{ wins } \mathsf{G}_1\right] \\
\ge& \frac{1}{2}\left(1 - \Pr\left[\mathbf{Adv} \text{ wins } \mathsf{PKE}^{\mathsf{IND-CCA}}\right] - \overbrace{\Pr_{\substack{(\mathsf{pk},\mathsf{sk})\leftarrow\mathsf{KeyGen}_{\mathsf{KEM}}(1^n)\\(\mathsf{ct}^*_{\mathsf{KEM}},k^*)\leftarrow\mathsf{Encaps}(\mathsf{pk})}}\left[\mathsf{Decaps}_{\mathsf{sk}}(\mathsf{ct}^*_{\mathsf{KEM}}) \ne k^*\right]}^{\le\,\mathsf{negl}(n)\ \ \text{(by the correctness of KEM)}}\right) + \frac{1}{2}\Pr\left[\mathbf{Adv} \text{ wins } \mathsf{G}_1\right] \\
& \hspace{20em} \text{(by Union Bound)} \\
\ge& \frac{1}{2} + \frac{1}{2}\left(\Pr\left[\mathbf{Adv} \text{ wins } \mathsf{G}_1\right] - \Pr\left[\mathbf{Adv} \text{ wins } \mathsf{PKE}^{\mathsf{IND-CCA}}\right]\right) - \frac{1}{2}\mathsf{negl}(n),
\end{aligned}
$$

which gives us

$$
\Pr\left[\mathbf{Adv} \text{ wins } \mathsf{G}_1\right] - \Pr\left[\mathbf{Adv} \text{ wins } \mathsf{PKE}^{\mathsf{IND-CCA}}\right] \le 2\nu_1(n) + \mathsf{negl}(n) = \nu'_1(n), \quad \text{for some } \nu'_1(n) = \mathsf{negl}(n). \tag{4.4}
$$

Now consider the "complement" PPT adversary R' of R that outputs the exact opposite bit as R, i.e., $R'^{\mathsf{Decaps}_{\mathsf{sk}}(\cdot)}(\mathsf{pk}, \mathsf{ct}^*_{\mathsf{KEM}}, k^*) \triangleq 1 - R^{\mathsf{Decaps}_{\mathsf{sk}}(\cdot)}(\mathsf{pk}, \mathsf{ct}^*_{\mathsf{KEM}}, k^*)$. Likewise, since KEM is an IND-CCA-secure KEM and R' is a PPT adversary, by a similar reasoning, we deduce that there also exists a negligible function $\nu_2(\cdot)$ such that for all $n \in \mathbb{N}$

$$
\Pr\left[\mathbf{Adv} \text{ wins } \mathsf{G}_1\right] - \Pr\left[\mathbf{Adv} \text{ wins } \mathsf{PKE}^{\mathsf{IND-CCA}}\right] \ge -2\nu_2(n) - \mathsf{negl}(n) = -\nu'_2(n), \quad \text{for some } \nu'_2(n) = \mathsf{negl}(n). \tag{4.5}
$$

Let $\nu(n) \triangleq \max(\nu'_1(n), \nu'_2(n))$, which is clearly negligible. Combing (4.4) and (4.5), we finally obtain

$$
\left|\Pr\left[\mathbf{Adv} \text{ wins } \mathsf{G}_1\right] - \Pr\left[\mathbf{Adv} \text{ wins } \mathsf{PKE}^{\mathsf{IND-CCA}}\right]\right| \le \nu(n),
$$

as desired. ∎

Claim 4.2 and equation (4.3) together tell us that

$$\Pr\left[\mathbf{Adv}\text{ wins } \mathsf{G}_1\right] \geq \Pr\left[\mathbf{Adv}\text{ wins } \mathsf{PKE}^{\mathsf{IND-CCA}}\right] - \mathsf{negl}(n) = \frac{1}{2} + \varepsilon(n) - \mathsf{negl}(n). \tag{4.6}$$

This allows us to construct a PPT oracle adversary (a reduction) R that breaks the IND-CCA security of SKE. The construction of R is provided in the following IND-CCA experiment $\mathsf{SKE}^{\mathsf{IND-CCA}}$ of SKE. (Note that $k^* \leftarrow \mathsf{KeyGen}_{\mathsf{SKE}}(1^n)$ is equivalent to $k^* \xleftarrow{\$} \{0,1\}^{\ell(n)}$.)

Ch		$R^{\mathsf{Enc}^{\mathsf{oracle}}_{\mathsf{SKE}}(k^*,\cdot),\mathsf{Dec}^{\mathsf{oracle}}_{\mathsf{SKE}}(k^*,\cdot)}$		$\mathbf{Adv}^{\mathsf{Dec}_{\mathsf{PKE}}(\mathsf{sk},\cdot)}$
$k^* \xleftarrow{\$} \{0,1\}^{\ell(n)}$	$\xrightarrow{1^n}$	$(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{KeyGen}_{\mathsf{KEM}}(1^n)$		
		$(\mathsf{ct}^*_{\mathsf{KEM}},\cdot) \leftarrow \mathsf{Encaps}(\mathsf{pk})$	$\xrightarrow{\mathsf{pk}}$	
	$\xleftarrow{\mathsf{ct}_{\mathsf{SKE}}}$ $\xrightarrow{\mathsf{Dec}_{\mathsf{SKE}}(k^*,\mathsf{ct}_{\mathsf{SKE}})}$	$\underline{\mathsf{Dec}_{\mathsf{PKE}}(\mathsf{sk},\mathsf{ct}_{\mathsf{PKE}})}$ 1: Parse $(\mathsf{ct}_{\mathsf{KEM}},\mathsf{ct}_{\mathsf{SKE}}) \coloneqq \mathsf{ct}_{\mathsf{PKE}}$ 2: **if** $\mathsf{ct}_{\mathsf{KEM}} = \mathsf{ct}^*_{\mathsf{KEM}}$ **then** 3: **return** $\mathsf{Dec}^{\mathsf{oracle}}_{\mathsf{SKE}}(k^*,\mathsf{ct}_{\mathsf{SKE}})$ 4: **else** 5: $k \coloneqq \mathsf{Decaps}_{\mathsf{sk}}(\mathsf{ct}_{\mathsf{KEM}})$ 6: **return** $\begin{cases} \perp & \text{if } k = \perp \\ \mathsf{Dec}_{\mathsf{SKE}}(k,\mathsf{ct}_{\mathsf{SKE}}) & \text{if } k \neq \perp \end{cases}$	$\xleftarrow{\mathsf{ct}_{\mathsf{PKE}\,i}}$ $\xrightarrow{\mathsf{Dec}_{\mathsf{PKE}}(\mathsf{sk},\mathsf{ct}_{\mathsf{PKE}\,i})}$	Query $\mathsf{Dec}_{\mathsf{PKE}}(\mathsf{sk},\cdot)$
$b \xleftarrow{\$} \{0,1\}$	$\xleftarrow{(m_0,m_1)}$		$\xleftarrow{(m_0,m_1)}$	
$\mathsf{ct}^*_{\mathsf{SKE}} \leftarrow \mathsf{Enc}_{\mathsf{SKE}}(k^*,m_b)$	$\xrightarrow{\mathsf{ct}^*_{\mathsf{SKE}}}$	$\mathsf{ct}^*_{\mathsf{PKE}} \coloneqq (\mathsf{ct}^*_{\mathsf{KEM}},\mathsf{ct}^*_{\mathsf{SKE}})$	$\xrightarrow{\mathsf{ct}^*_{\mathsf{PKE}}}$	
	$\xleftarrow{\mathsf{ct}_{\mathsf{SKE}}\neq\mathsf{ct}^*_{\mathsf{SKE}}}$ $\xrightarrow{\mathsf{Dec}^{\mathsf{oracle}}_{\mathsf{SKE}}(k^*,\mathsf{ct}_{\mathsf{SKE}})}$	$\underline{\mathsf{Dec}_{\mathsf{PKE}}(\mathsf{sk},\mathsf{ct}_{\mathsf{PKE}} \neq \mathsf{ct}^*_{\mathsf{PKE}})}$ 1: Parse $(\mathsf{ct}_{\mathsf{KEM}},\mathsf{ct}_{\mathsf{SKE}}) \coloneqq \mathsf{ct}_{\mathsf{PKE}}$ 2: **if** $\mathsf{ct}_{\mathsf{KEM}} = \mathsf{ct}^*_{\mathsf{KEM}}$ **then** 3: **return** $\mathsf{Dec}^{\mathsf{oracle}}_{\mathsf{SKE}}(k^*,\mathsf{ct}_{\mathsf{SKE}})$ 4: **else** 5: $k \coloneqq \mathsf{Decaps}_{\mathsf{sk}}(\mathsf{ct}_{\mathsf{KEM}})$ 6: **return** $\begin{cases} \perp & \text{if } k = \perp \\ \mathsf{Dec}_{\mathsf{SKE}}(k,\mathsf{ct}_{\mathsf{SKE}}) & \text{if } k \neq \perp \end{cases}$	$\xleftarrow{\mathsf{ct}_{\mathsf{PKE}\,i}\neq\mathsf{ct}^*_{\mathsf{PKE}}}$ $\xrightarrow{\mathsf{Dec}_{\mathsf{PKE}}(\mathsf{sk},\mathsf{ct}_{\mathsf{PKE}\,i})}$	Query $\mathsf{Dec}_{\mathsf{PKE}}(\mathsf{sk},\cdot)$
R wins if $b' = b$	$\xleftarrow{b'}$		$\xleftarrow{b'}$	

Figure 11: Experiment $\mathsf{SKE}^{\mathsf{IND-CCA}}$

Clearly R runs in PPT as $\mathbf{Adv}$ also runs in PPT.

Observe that the *view* of $\mathbf{Adv}$ in experiment $\mathsf{SKE}^{\mathsf{IND-CCA}}$ is distributed identically to that of $\mathbf{Adv}$ in game G_1. This is because in $\mathsf{SKE}^{\mathsf{IND-CCA}}$, we have just delegated the job of challenger in game G_1 to the new challenger and PPT oracle adversary R. To make it clearer, it's not hard to check that the decryption oracle $\mathsf{Dec}_{\mathsf{PKE}}(\mathsf{sk},\cdot)$ simulated by R here is in fact equivalent to the one in G_1. Therefore, for all $n \in \mathbb{N}$, the win rate of R in experiment $\mathsf{SKE}^{\mathsf{IND-CCA}}$ is simply

$$\Pr\left[R\text{ wins } \mathsf{KEM}^{\mathsf{IND-CCA}}\right] = \Pr\left[\mathbf{Adv}\text{ wins } \mathsf{G}_1\right] \overset{(4.6)}{\geq} = \frac{1}{2} + \varepsilon(n) - \mathsf{negl}(n).$$

This tells us that the advantage of R is at least $\varepsilon(n) - \mathsf{negl}(n)$, which is nonnegligible.

Hence, by definition, $\mathsf{SKE} = (\mathsf{KeyGen}_{\mathsf{SKE}},\mathsf{Enc}_{\mathsf{SKE}},\mathsf{Dec}_{\mathsf{SKE}})$ is not IND-CCA-secure, which contradicts our assumption, so $\mathsf{PKE} = (\mathsf{KeyGen}_{\mathsf{PKE}},\mathsf{Enc}_{\mathsf{PKE}},\mathsf{Dec}_{\mathsf{PKE}})$ must be IND-CCA-secure. ⊛

Problem 5 (OR-Sigma protocol is secure).

Suppose that we have two Sigma protocols (P_0, V_0) for $\mathcal{R}_0 \subseteq \mathcal{X}_0 \times \mathcal{Y}_0$ and (P_1, V_1) for $\mathcal{R}_1 \subseteq \mathcal{X}_1 \times \mathcal{Y}_1$. We assume that

- Both Sigma protocols use the same challenge space $\mathcal{C} = \{0,1\}^n$.
- Both protocols are HVZK with simulators Sim_0 and Sim_1 respectively.

We can combine them to form a Sigma protocol for the relation $\mathcal{R}_{\mathsf{OR}} \subseteq \mathcal{X}_{\mathsf{OR}} \times \mathcal{Y}_{\mathsf{OR}}$, where $\mathcal{X}_{\mathsf{OR}} \triangleq \mathcal{X}_0 \times \mathcal{X}_1$ and $\mathcal{Y}_{\mathsf{OR}} \triangleq \{0,1\} \times (\mathcal{Y}_0 \cup \mathcal{Y}_1)$:

$$\mathcal{R}_{\mathsf{OR}} \triangleq \left\{ ((x_0, x_1), (b, y)) \in \mathcal{X}_{\mathsf{OR}} \times \mathcal{Y}_{\mathsf{OR}} \;\middle|\; (x_b, y) \in \mathcal{R}_b \right\}.$$

In other words, for a given pair of statements $x_0 \in \mathcal{X}_0$ and $x_1 \in \mathcal{X}_1$, this OR protocol allows a prover to convince a skeptical verifier that he “knows” a witness for x_0 or a witness for x_1. Nothing else should be revealed. In particular the protocol should not reveal which witness the prover actually have. The protocol (P, V) runs as follows, where the prover P is initialized with $((x_0, x_1), (b, y)) \in \mathcal{R}_{\mathsf{OR}}$, the verifier V is initialized with $(x_0, x_1) \in \mathcal{X}_0 \times \mathcal{X}_1$, and $d \triangleq 1 - b$:

(a) P computes $(\mathsf{com}_d, \mathsf{ch}_d, \mathsf{resp}_d) \leftarrow \mathsf{Sim}_d(x_d)$. P also runs $P_b(x_b, y)$ to get a commitment com_b and sends the pair $(\mathsf{com}_0, \mathsf{com}_1)$ to V.

(b) V computes $\mathsf{ch} \leftarrow \mathcal{C}$ and sends the challenge ch to P.

(c) P computes $\mathsf{ch}_b := \mathsf{ch} \oplus \mathsf{ch}_d$. P feeds the challenge ch_b to $P_b(x_b, y)$, obtaining a response resp_b, and sends $(\mathsf{ch}_0, \mathsf{resp}_0, \mathsf{resp}_1)$ to V.

(d) V computes $\mathsf{ch}_1 := \mathsf{ch} \oplus \mathsf{ch}_0$ and checks that $(\mathsf{com}_i, \mathsf{ch}_i, \mathsf{resp}_i)$ is an accepting conversation for x_i, where $i = 0, 1$.

Show that the above protocol is indeed a Sigma protocol for $\mathcal{R}_{\mathsf{OR}}$. In other words, show that the protocol satisfies completeness, special soundness, and HVZK.

Answer.

Proof of Completeness.

Let $((x_0, x_1), (b, y)) \in \mathcal{R}_{\mathsf{OR}}$.

For simplicity, for any Sigma protocol (P, V) for an NP relation $\mathcal{R} \subseteq \mathcal{X} \times \mathcal{Y}$, we define the last step of V as a deterministic function (or rather, a predicate) $V(x, \mathsf{com}, \mathsf{ch}, \mathsf{resp}) \in \{0,1\}$ (using the same symbol V as the verifier), where $x \in \mathcal{X}$ is the statement given to the verifier, and $(\mathsf{com}, \mathsf{ch}, \mathsf{resp})$ the transcript of a conversation. We also define a transcript $(\mathsf{com}, \mathsf{ch}, \mathsf{resp})$ of a conversation in a Sigma protocol (P, V) to be *accepting* for a statement $x \in \mathcal{X}$ if $V(x, \mathsf{com}, \mathsf{ch}, \mathsf{resp}) = 1$.

Since (P_0, V_0) and (P_1, V_1) are both Sigma protocols and are thus complete, for each $b \in \{0,1\}$, there exists $\varepsilon_b(n) = \mathsf{negl}(n)$ such that for every $n \in \mathbb{N}$ and every $(x, y) \in \mathcal{R}_b$,

$$1 - \varepsilon_b(n) = \Pr\left[\langle P_b(x, y), V_b(x)\rangle = 1\right] = \Pr_{\substack{(\mathsf{com},\mathsf{st}) \leftarrow P_b(x,y) \\ \mathsf{ch} \xleftarrow{\$} \{0,1\}^n \\ \mathsf{resp} \leftarrow P_b(\mathsf{st},\mathsf{ch})}} \left[V_b(x, \mathsf{com}, \mathsf{ch}, \mathsf{resp}) = 1\right], \tag{5.1}$$

where st denotes the internal state of P_b.

Also, since (P_0, V_0) and (P_1, V_1) are both HVZK with respective simulators Sim_0 and Sim_1, for each $b \in \{0,1\}$, every $n \in \mathbb{N}$, and every $(x, y) \in \mathcal{R}_b$, we have

$$(\mathsf{com}', \mathsf{ch}', \mathsf{resp}') \stackrel{\mathrm{d}}{=} (\mathsf{com}, \mathsf{ch}, \mathsf{resp}) \qquad \text{where} \qquad \frac{(\mathsf{com}', \mathsf{ch}', \mathsf{resp}') \leftarrow \mathsf{Sim}_b(x)}{(\mathsf{com}, \mathsf{st}) \leftarrow P_b(x, y), \mathsf{ch} \xleftarrow{\$} \{0,1\}^n, \mathsf{resp} \leftarrow P_b(\mathsf{st}, \mathsf{ch})},$$

which implies that

$$\Pr[V_b(x, \mathsf{com}', \mathsf{ch}', \mathsf{resp}') = 1] = \sum_{(com, ch, resp)} \Pr[(\mathsf{com}', \mathsf{ch}', \mathsf{resp}') = (com, ch, resp)] \Pr[V_b(x, com, ch, resp) = 1]$$

$$= \sum_{(com, ch, resp)} \Pr[(\mathsf{com}, \mathsf{ch}, \mathsf{resp}) = (com, ch, resp)] \Pr[V_b(x, com, ch, resp) = 1] = \Pr[V_b(x, \mathsf{com}, \mathsf{ch}, \mathsf{resp}) = 1]$$

as $V_b(x, \cdot) : (com, ch, resp) \mapsto 0/1$ is simply a deterministic function of the transcript $(com, ch, resp)$ of a conversation. Then, by the completeness of (P_b, V_b) (cf. (5.1)), we have

$$\Pr[V_b(x, \mathsf{com}', \mathsf{ch}', \mathsf{resp}') = 1] = \Pr[V_b(x, \mathsf{com}, \mathsf{ch}, \mathsf{resp}) = 1] = 1 - \mathsf{negl}(n). \tag{5.2}$$

This means that the transcript $(\mathsf{com}', \mathsf{ch}', \mathsf{resp}')$ simulated by $\mathsf{Sim}_b(x)$ can be accepted by $V_b(x)$ with high probability, as if it were a transcript of a conversation between $P_b(x, y)$ and $V_b(x)$.

Finally, for all $n \in \mathbb{N}$ and every $((x_0, x_1), (b, y)) \in \mathcal{R}_{\mathsf{OR}}$, since $(x_b, y) \in \mathcal{R}_b$, we have

$$\begin{aligned}
&\Pr\left[\langle P((x_0, x_1), (b, y)), V(x_0, x_1)\rangle = 1\right] \\
&= \Pr\left[(\mathsf{com}_0, \mathsf{ch}_0, \mathsf{resp}_0) \text{ is accepting for } x_0 \text{ in } (P_0, V_0) \wedge (\mathsf{com}_1, \mathsf{ch}_1, \mathsf{resp}_1) \text{ is accepting for } x_1 \text{ in } (P_1, V_1)\right] \\
&= \Pr\left[V_0(x_0, \mathsf{com}_0, \mathsf{ch}_0, \mathsf{resp}_0) = 1 \wedge V_1(x_1, \mathsf{com}_1, \mathsf{ch}_1, \mathsf{resp}_1) = 1\right] \\
&= \Pr\left[V_b(x_b, \mathsf{com}_b, \mathsf{ch}_b, \mathsf{resp}_b) = 1 \wedge V_{1-b}(x_{1-b}, \mathsf{com}_{1-b}, \mathsf{ch}_{1-b}, \mathsf{resp}_{1-b}) = 1\right] \\
&= \Pr_{\substack{(\mathsf{com}_{1-b}, \mathsf{ch}_{1-b}, \mathsf{resp}_{1-b}) \leftarrow \mathsf{Sim}_{1-b}(x_{1-b}) \\ (\mathsf{com}_b, \mathsf{st}) \leftarrow P_b(x_b, y) \\ \mathsf{ch} \xleftarrow{\$} \{0,1\}^n \\ \mathsf{resp}_b \leftarrow P_b(\mathsf{st}, \mathsf{ch} \oplus \mathsf{ch}_{1-b})}} \left[V_b(x_b, \mathsf{com}_b, \mathsf{ch} \oplus \mathsf{ch}_{1-b}, \mathsf{resp}_b) = 1 \wedge V_{1-b}(x_{1-b}, \mathsf{com}_{1-b}, \mathsf{ch}_{1-b}, \mathsf{resp}_{1-b}) = 1\right] \\
&\geq 1 - \left(1 - \Pr_{\substack{(\cdot, \mathsf{ch}_{1-b}, \mathrm{cot}) \leftarrow \mathsf{Sim}_{1-b}(x_{1-b}) \\ (\mathsf{com}_b, \mathsf{st}) \leftarrow P_b(x_b, y) \\ \mathsf{ch} \xleftarrow{\$} \{0,1\}^n \\ \mathsf{resp}_b \leftarrow P_b(\mathsf{st}, \mathsf{ch} \oplus \mathsf{ch}_{1-b})}} [V_b(x_b, \mathsf{com}_b, \mathsf{ch} \oplus \mathsf{ch}_{1-b}, \mathsf{resp}_b) = 1]\right) \\
&\qquad - \left(1 - \Pr_{(\mathsf{com}_{1-b}, \mathsf{ch}_{1-b}, \mathsf{resp}_{1-b}) \leftarrow \mathsf{Sim}_{1-b}(x_{1-b})} \left[V_{1-b}(x_{1-b}, \mathsf{com}_{1-b}, \mathsf{ch}_{1-b}, \mathsf{resp}_{1-b}) = 1\right]\right) \qquad \text{(by Union Bound)} \\
&\overset{(\text{🍄})}{=} \overbrace{\Pr_{\substack{(\mathsf{com}_b, \mathsf{st}) \leftarrow P_b(x_b, y) \\ \mathsf{ch}_b \xleftarrow{\$} \{0,1\}^n \\ \mathsf{resp}_b \leftarrow P_b(\mathsf{st}, \mathsf{ch}_b)}} [V_b(x_b, \mathsf{com}_b, \mathsf{ch}_b, \mathsf{resp}_b) = 1]}^{= 1 - \mathsf{negl}(n) \ \text{(by (5.1) and } (x_b, y) \in \mathcal{R}_b)} \\
&\qquad \underbrace{+ \Pr_{(\mathsf{com}_{1-b}, \mathsf{ch}_{1-b}, \mathsf{resp}_{1-b}) \leftarrow \mathsf{Sim}_{1-b}(x_{1-b})} \left[V_{1-b}(x_{1-b}, \mathsf{com}_{1-b}, \mathsf{ch}_{1-b}, \mathsf{resp}_{1-b}) = 1\right]}_{= 1 - \mathsf{negl}(n) \ \text{(by (5.2))}} - 1 \\
&= 1 - \mathsf{negl}(n) - \mathsf{negl}(n) = 1 - \mathsf{negl}(n).
\end{aligned}$$

Note that in the step (🍄), since $\mathsf{ch} \xleftarrow{\$} \{0,1\}^n$ is uniformly random and independent of ch_d, $\mathsf{ch}_b \triangleq \mathsf{ch} \oplus \mathsf{ch}_d$ must be uniformly random and independent of ch_d. Thus, we substitute every occurrence of $\mathsf{ch} \oplus \mathsf{ch}_d$ with a shared, fresh $\mathsf{ch}_b \xleftarrow{\$} \{0,1\}^n$ (and every occurrence of ch_d with $\mathsf{ch} \oplus \mathsf{ch}_b$).

Hence, by definition, the (potentially) Sigma protocol (P, V) for $\mathcal{R}_{\mathsf{OR}}$ satisfies completeness.

Proof of Special Soundness.

Since (P_0, V_0) and (P_1, V_1) are both Sigma protocols and are thus specially sound, there exist polynomial-time algorithms $\mathsf{Ext}_0, \mathsf{Ext}_1$, each of which takes as input $x_b \in \mathcal{X}_b$ and two accepting conversation transcripts $(\mathsf{com}_b, \mathsf{ch}_b, \mathsf{resp}_b)$ and $(\mathsf{com}_b, \mathsf{ch}'_b, \mathsf{resp}'_b)$ where $\mathsf{ch}_b \neq \mathsf{ch}'_b$, and outputs $y_b \in \mathcal{Y}_b$ such that $(x_b, y_b) \in \mathcal{R}_b$. That is, for each $b \in \{0,1\}$, for any $x_b \in \mathcal{X}_b$ and any conversation transcripts $(\mathsf{com}_b, \mathsf{ch}_b, \mathsf{resp}_b)$ and $(\mathsf{com}_b, \mathsf{ch}'_b, \mathsf{resp}'_b)$, there is

$$\mathsf{ch}_b \neq \mathsf{ch}'_b \wedge V_b(x_b, \mathsf{com}_b, \mathsf{ch}_b, \mathsf{resp}_b) = V_b(x_b, \mathsf{com}_b, \mathsf{ch}'_b, \mathsf{resp}'_b) = 1 \implies \big(x_b, \mathsf{Ext}_b(\mathsf{com}_b, \mathsf{ch}_b, \mathsf{ch}'_b, \mathsf{resp}_b, \mathsf{resp}'_b)\big) \in \mathcal{R}_b. \tag{5.3}$$

We construct as follows a polynomial-time deterministic algorithm Ext that takes as input a statement $(x_0, x_1) \in \mathcal{X}_{\mathsf{OR}} \triangleq \mathcal{X}_0 \times \mathcal{X}_1$ and two accepting conversation transcripts $(\mathsf{com}, \mathsf{ch}, \mathsf{resp})$ and $(\mathsf{com}, \mathsf{ch}', \mathsf{resp}')$ where $\mathsf{ch} \neq \mathsf{ch}'$, and outputs $(b, y) \in \mathcal{Y}_{\mathsf{OR}} \triangleq \{0,1\} \times (\mathcal{Y}_0 \cup \mathcal{Y}_1)$ such that $((x_0, x_1), (b, y)) \in \mathcal{R}_{\mathsf{OR}}$:

<u>$\mathsf{Ext}\left((x_0, x_1) \in \mathcal{X}_{\mathsf{OR}}, \mathsf{com}, \mathsf{ch}, \mathsf{ch}', \mathsf{resp}, \mathsf{resp}'\right)$</u>

Parse $(\mathsf{com}_0, \mathsf{com}_1) := \mathsf{com}$, $(\mathsf{ch}_0, \mathsf{resp}_0, \mathsf{resp}_1) := \mathsf{resp}$, $(\mathsf{ch}'_0, \mathsf{resp}'_0, \mathsf{resp}'_1) := \mathsf{resp}'$
Compute $\mathsf{ch}_1 := \mathsf{ch} \oplus \mathsf{ch}_0$, $\mathsf{ch}'_1 := \mathsf{ch}' \oplus \mathsf{ch}'_0$
if $\mathsf{ch}_0 \neq \mathsf{ch}'_0$ **then**
    $(b, y) := (0, \mathsf{Ext}_0(x_0, \mathsf{com}_0, \mathsf{ch}_0, \mathsf{ch}'_0, \mathsf{resp}_0, \mathsf{resp}'_0))$
else
    $(b, y) := (1, \mathsf{Ext}_1(x_1, \mathsf{com}_1, \mathsf{ch}_1, \mathsf{ch}'_1, \mathsf{resp}_1, \mathsf{resp}'_1))$
return (b, y)

Clearly Ext is deterministic and polynomial-time as both Ext_0 and Ext_1 are deterministic and polynomial-time. Then, using Hoare Logic, we verify that for any $(x_0, x_1) \in \mathcal{X}_{\mathsf{OR}}$ and any conversation transcripts $(\mathsf{com}, \mathsf{ch}, \mathsf{resp})$ and $(\mathsf{com}, \mathsf{ch}', \mathsf{resp}')$ where $\mathsf{ch} \neq \mathsf{ch}'$ such that both transcripts are accepting, i.e., $V((x_0, x_1), \mathsf{com}, \mathsf{ch}, \mathsf{resp}) = V((x_0, x_1), \mathsf{com}, \mathsf{ch}', \mathsf{resp}') = 1$, there is

$$\big((x_0, x_1), (b, \mathsf{Ext}(\mathsf{com}, \mathsf{ch}, \mathsf{ch}', \mathsf{resp}, \mathsf{resp}'))\big) \in \mathcal{R}_{\mathsf{OR}}.$$

Observation 5.1. For any $\mathsf{ch}, \mathsf{ch}', \mathsf{ch}_0, \mathsf{ch}'_0 \in \{0,1\}^n$, if $\mathsf{ch} \neq \mathsf{ch}'$, then either $\mathsf{ch}_0 \neq \mathsf{ch}'_0$ or $\mathsf{ch}_0 = \mathsf{ch}'_0$ holds, the latter of which implies $\mathsf{ch} \oplus \mathsf{ch}_0 \neq \mathsf{ch}' \oplus \mathsf{ch}'_0$. That is,

$$\vDash \forall \mathsf{ch}, \mathsf{ch}', \mathsf{ch}_0, \mathsf{ch}'_0 \in \{0,1\}^n \Big(\mathsf{ch} \neq \mathsf{ch}' \implies \mathsf{ch}_0 \neq \mathsf{ch}'_0 \vee \mathsf{ch} \oplus \mathsf{ch}_0 \neq \mathsf{ch}' \oplus \mathsf{ch}'_0\Big).$$

$\underline{\mathsf{Ext}\,((x_0,x_1)\in\mathcal{X}_{\mathsf{OR}},\mathsf{com},\mathsf{ch},\mathsf{ch}',\mathsf{resp},\mathsf{resp}')}$

$\{\mathsf{ch}\neq\mathsf{ch}'\wedge V((x_0,x_1),\mathsf{com},\mathsf{ch},\mathsf{resp})=V((x_0,x_1),\mathsf{com},\mathsf{ch}',\mathsf{resp}')=1\}$ ▷ Precondition

$\{(\mathsf{ch}_0\neq\mathsf{ch}_0'\vee\mathsf{ch}\oplus\mathsf{ch}_0\neq\mathsf{ch}'\oplus\mathsf{ch}_0')\wedge V((x_0,x_1),\mathsf{com},\mathsf{ch},\mathsf{resp})=V((x_0,x_1),\mathsf{com},\mathsf{ch}',\mathsf{resp}')=1\}$ ▷ Implied by Observation 5.1

Parse $(\mathsf{com}_0,\mathsf{com}_1):=\mathsf{com}$, $(\mathsf{ch}_0,\mathsf{resp}_0,\mathsf{resp}_1):=\mathsf{resp}$, $(\mathsf{ch}_0',\mathsf{resp}_0',\mathsf{resp}_1'):=\mathsf{resp}'$

$$\left\{(\mathsf{ch}_0\neq\mathsf{ch}_0'\vee\mathsf{ch}\oplus\mathsf{ch}_0\neq\mathsf{ch}'\oplus\mathsf{ch}_0')\wedge\begin{array}{r}V((x_0,x_1),(\mathsf{com}_0,\mathsf{com}_1),\mathsf{ch},(\mathsf{ch}_0,\mathsf{resp}_0,\mathsf{resp}_1))=\\V((x_0,x_1),(\mathsf{com}_0,\mathsf{com}_1),\mathsf{ch}',(\mathsf{ch}_0',\mathsf{resp}_0',\mathsf{resp}_1'))=1\end{array}\right\}$$

$$\left\{(\mathsf{ch}_0\neq\mathsf{ch}_0'\vee\mathsf{ch}\oplus\mathsf{ch}_0\neq\mathsf{ch}'\oplus\mathsf{ch}_0')\wedge\begin{array}{r}V_0(x_0,\mathsf{com}_0,\mathsf{ch}_0,\mathsf{resp}_0)=V_1(x_1,\mathsf{com}_1,\mathsf{ch}\oplus\mathsf{ch}_0,\mathsf{resp}_1)=\\V_0(x_0,\mathsf{com}_0,\mathsf{ch}_0',\mathsf{resp}_0')=V_1(x_1,\mathsf{com}_1,\mathsf{ch}'\oplus\mathsf{ch}_0',\mathsf{resp}_1')=1\end{array}\right\}$$

▷ Implied by def. of V

Compute $\mathsf{ch}_1:=\mathsf{ch}\oplus\mathsf{ch}_0$, $\mathsf{ch}_1':=\mathsf{ch}'\oplus\mathsf{ch}_0'$

$$\left\{(\mathsf{ch}_0\neq\mathsf{ch}_0'\vee\mathsf{ch}_1\neq\mathsf{ch}_1')\wedge\begin{array}{r}V_0(x_0,\mathsf{com}_0,\mathsf{ch}_0,\mathsf{resp}_0)=V_1(x_1,\mathsf{com}_1,\mathsf{ch}_1,\mathsf{resp}_1)=\\V_0(x_0,\mathsf{com}_0,\mathsf{ch}_0',\mathsf{resp}_0')=V_1(x_1,\mathsf{com}_1,\mathsf{ch}_1',\mathsf{resp}_1')=1\end{array}\right\}$$

if $\mathsf{ch}_0\neq\mathsf{ch}_0'$ **then**

$$\left\{(\mathsf{ch}_0\neq\mathsf{ch}_0'\vee\mathsf{ch}_1\neq\mathsf{ch}_1')\wedge\begin{array}{r}V_0(x_0,\mathsf{com}_0,\mathsf{ch}_0,\mathsf{resp}_0)=V_1(x_1,\mathsf{com}_1,\mathsf{ch}_1,\mathsf{resp}_1)=\\V_0(x_0,\mathsf{com}_0,\mathsf{ch}_0',\mathsf{resp}_0')=V_1(x_1,\mathsf{com}_1,\mathsf{ch}_1',\mathsf{resp}_1')=1\end{array}\wedge\mathsf{ch}_0\neq\mathsf{ch}_0'\right\}$$

$\{\mathsf{ch}_0\neq\mathsf{ch}_0'\wedge V_0(x_0,\mathsf{com}_0,\mathsf{ch}_0,\mathsf{resp}_0)=V_0(x_0,\mathsf{com}_0,\mathsf{ch}_0',\mathsf{resp}_0')=1\}$ ▷ Implied

$\{(x_0,\mathsf{Ext}_0(x_0,\mathsf{com}_0,\mathsf{ch}_0,\mathsf{ch}_0',\mathsf{resp}_0,\mathsf{resp}_0'))\in\mathcal{R}_0\}$ ▷ Implied by (5.3)

$(b,y):=(0,\mathsf{Ext}_0(x_0,\mathsf{com}_0,\mathsf{ch}_0,\mathsf{ch}_0',\mathsf{resp}_0,\mathsf{resp}_0'))$

$\{(x_b,y)\in\mathcal{R}_b\}$

else

$$\left\{(\mathsf{ch}_0\neq\mathsf{ch}_0'\vee\mathsf{ch}_1\neq\mathsf{ch}_1')\wedge\begin{array}{r}V_0(x_0,\mathsf{com}_0,\mathsf{ch}_0,\mathsf{resp}_0)=V_1(x_1,\mathsf{com}_1,\mathsf{ch}_1,\mathsf{resp}_1)=\\V_0(x_0,\mathsf{com}_0,\mathsf{ch}_0',\mathsf{resp}_0')=V_1(x_1,\mathsf{com}_1,\mathsf{ch}_1',\mathsf{resp}_1')=1\end{array}\wedge\mathsf{ch}_0=\mathsf{ch}_0'\right\}$$

$\{\mathsf{ch}_1\neq\mathsf{ch}_1'\wedge V_1(x_1,\mathsf{com}_1,\mathsf{ch}_1,\mathsf{resp}_1)=V_1(x_1,\mathsf{com}_1,\mathsf{ch}_1',\mathsf{resp}_1')=1\}$ ▷ Implied

$\{(x_1,\mathsf{Ext}_1(x_1,\mathsf{com}_1,\mathsf{ch}_1,\mathsf{ch}_1',\mathsf{resp}_1,\mathsf{resp}_1'))\in\mathcal{R}_1\}$ ▷ Implied by (5.3)

$(b,y):=(1,\mathsf{Ext}_1(x_1,\mathsf{com}_1,\mathsf{ch}_1,\mathsf{ch}_1',\mathsf{resp}_1,\mathsf{resp}_1'))$

$\{(x_b,y)\in\mathcal{R}_b\}$

$\{(x_b,y)\in\mathcal{R}_b\}$

$\{((x_0,x_1),(b,y))\in\mathcal{R}_{\mathsf{OR}}\}$ ▷ Implied by def. of $\mathcal{R}_{\mathsf{OR}}$ (Postcondition)

return (b,y)

Therefore, by definition, the (potentially) Sigma protocol (P,V) for $\mathcal{R}_{\mathsf{OR}}$ satisfies special soundness.

Proof of HVZK.

As reasoned before, since (P_0,V_0) and (P_1,V_1) are both HVZK with respective simulators Sim_0 and Sim_1, for each $b\in\{0,1\}$ and $n\in\mathbb{N}$, for every $(x_b,y_b)\in\mathcal{R}_b$, we have

$$(\mathsf{com}_b',\mathsf{ch}_b',\mathsf{resp}_b')\overset{\mathrm{d}}{=}(\mathsf{com}_b,\mathsf{ch}_b,\mathsf{resp}_b)\qquad\text{where}\qquad\begin{array}{c}(\mathsf{com}_b',\mathsf{ch}_b',\mathsf{resp}_b')\leftarrow\mathsf{Sim}_b(x_b)\\(\mathsf{com}_b,\mathsf{st})\leftarrow P_b(x_b,y_b),\mathsf{ch}_b\overset{\$}{\leftarrow}\{0,1\}^n,\mathsf{resp}_b\leftarrow P_b(\mathsf{st},\mathsf{ch}_b)\end{array}.\tag{5.4}$$

We construct as follows a PPT algorithm Sim such that for all $n\in\mathbb{N}$ and $((x_0,x_1),(b,y))\in\mathcal{R}_{\mathsf{OR}}$, on input (x_0,x_1), Sim outputs a transcript $(\mathsf{com}',\mathsf{ch}',\mathsf{resp}')$ of a (simulated) conversation such that $(\mathsf{com}',\mathsf{ch}',\mathsf{resp}')$ is distributed identically to a transcript $(\mathsf{com},\mathsf{ch},\mathsf{resp})$ of a conversation between

$P((x_0,x_1),(b,y))$ and $V(x_0,x_1)$, i.e.,

$$(\mathsf{com}',\mathsf{ch}',\mathsf{resp}') \overset{\mathrm{d}}{\triangleq} (\mathsf{com},\mathsf{ch},\mathsf{resp}) \qquad \text{where} \qquad \begin{array}{c} (\mathsf{com}',\mathsf{ch}',\mathsf{resp}') \leftarrow \mathsf{Sim}(x_0,x_1) \\ (\mathsf{com},\mathsf{st}) \leftarrow P((x_0,x_1),(b,y)), \mathsf{ch} \xleftarrow{\$} \{0,1\}^n, \mathsf{resp} \leftarrow P(\mathsf{st},\mathsf{ch}) \end{array}. \tag{5.5}$$

$\underline{\mathsf{Sim}\,((x_0,x_1)\in\mathcal{X}_{\mathsf{OR}})}$

1: $(\mathsf{com}'_0,\mathsf{ch}'_0,\mathsf{resp}'_0) \leftarrow \mathsf{Sim}_0(x_0)$

2: $(\mathsf{com}'_1,\mathsf{ch}'_1,\mathsf{resp}'_1) \leftarrow \mathsf{Sim}_1(x_1)$

3: **return** $(\mathsf{com}' := (\mathsf{com}'_0,\mathsf{com}'_1), \mathsf{ch}' := \mathsf{ch}'_0 \oplus \mathsf{ch}'_1, \mathsf{resp}' := (\mathsf{ch}'_0,\mathsf{resp}'_0,\mathsf{resp}'_1))$

Clearly Sim is a PPT algorithm. We then prove (5.5) as follows. For all $n\in\mathbb{N}$ and $((x_0,x_1),(b,y))\in\mathcal{R}_{\mathsf{OR}}$, define the following random variables:

- $(\mathsf{com} = (\mathsf{com}_0,\mathsf{com}_1),\mathsf{st}) \leftarrow P((x_0,x_1),(b,y))$, $\mathsf{ch} \xleftarrow{\$} \{0,1\}^n$, $\mathsf{resp} = (\mathsf{ch}_0,\mathsf{resp}_0,\mathsf{resp}_1) \leftarrow P(\mathsf{st},\mathsf{ch})$, or equivalently,
  - $(\mathsf{com}_{1-b},\mathsf{ch}_{1-b},\mathsf{resp}_{1-b}) \leftarrow \mathsf{Sim}_{1-b}(x_{1-b})$,
  - $(\mathsf{com}_b,\mathsf{st}) \leftarrow P_b(x_b,y)$,
  - $\mathsf{ch} \xleftarrow{\$} \{0,1\}^n$,
  - $\mathsf{ch}_b \triangleq \mathsf{ch}\oplus\mathsf{ch}_{1-b}$, and
  - $\mathsf{resp}_b \leftarrow P_b(\mathsf{st},\mathsf{ch}_b)$;
- $(\mathsf{com}' = (\mathsf{com}'_0,\mathsf{com}'_1), \mathsf{ch}' = \mathsf{ch}'_0\oplus\mathsf{ch}'_1, \mathsf{resp}' = (\mathsf{ch}'_0,\mathsf{resp}'_0,\mathsf{resp}'_1)) \leftarrow \mathsf{Sim}(x_0,x_1)$, or equivalently,
  - $(\mathsf{com}'_0,\mathsf{ch}'_0,\mathsf{resp}'_0) \leftarrow \mathsf{Sim}_0(x_0)$ and
  - $(\mathsf{com}'_1,\mathsf{ch}'_1,\mathsf{resp}'_1) \leftarrow \mathsf{Sim}_1(x_1)$.

Then, to arrive at (5.5), there are two cases to show:

- **Case 1**: $b=0$. Then we have

$$\begin{aligned}
&(\mathsf{com}',\mathsf{ch}',\mathsf{resp}') \\
&= \left((\mathsf{com}'_0,\mathsf{com}'_1), \mathsf{ch}'_0\oplus\mathsf{ch}'_1, (\mathsf{ch}'_0,\mathsf{resp}'_0,\mathsf{resp}'_1)\right) \\
&= \left((\mathsf{com}'_b,\mathsf{com}'_{1-b}), \mathsf{ch}'_b\oplus\mathsf{ch}'_{1-b}, (\mathsf{ch}'_b,\mathsf{resp}'_b,\mathsf{resp}'_{1-b})\right) \\
&\overset{\mathrm{d}}{\triangleq} \left((\mathsf{com}'_b,\mathsf{com}_{1-b}), \mathsf{ch}'_b\oplus\mathsf{ch}_{1-b}, (\mathsf{ch}'_b,\mathsf{resp}'_b,\mathsf{resp}_{1-b})\right) \\
&\qquad (\because (\mathsf{com}_{1-b},\mathsf{ch}_{1-b},\mathsf{resp}_{1-b}), (\mathsf{com}'_{1-b},\mathsf{ch}'_{1-b},\mathsf{resp}'_{1-b}) \overset{\text{iid}}{\sim} \mathsf{Sim}_{1-b}(x_{1-b})) \\
&\overset{\mathrm{d}}{\triangleq} \left((com,\mathsf{com}_{1-b}), ch\oplus\mathsf{ch}_{1-b}, (ch,resp,\mathsf{resp}_{1-b})\right) \text{ where } (com,st)\leftarrow P_b(x_b,y), ch\xleftarrow{\$}\{0,1\}^n, resp\leftarrow P_b(st,ch) \\
&\qquad (\because (\mathsf{com}'_b,\mathsf{ch}'_b,\mathsf{resp}'_b) \overset{\mathrm{d}}{\triangleq} (com,ch,resp) \text{ by } (5.4)) \\
&\overset{\mathrm{d}}{\triangleq} \left((com,\mathsf{com}_{1-b}), ch^\star, (ch^\star\oplus\mathsf{ch}_{1-b}, resp, \mathsf{resp}_{1-b})\right) \\
&\qquad\qquad\qquad\qquad \text{where } (com,st)\leftarrow P_b(x_b,y), ch^\star\xleftarrow{\$}\{0,1\}^n, resp\leftarrow P_b(st,ch^\star\oplus\mathsf{ch}_{1-b}) \\
&\quad (\because ch \text{ is fresh, uniform, and independent of } \mathsf{ch}_{1-b} \implies ch^\star \triangleq ch\oplus\mathsf{ch}_{1-b} \text{ is also uniform,} \\
&\qquad \text{so we redefine } ch^\star \xleftarrow{\$} \{0,1\}^n, ch\triangleq ch^\star\oplus\mathsf{ch}_{1-b} \text{ so that } ch^\star \text{ becomes fresh}) \\
&\overset{\mathrm{d}}{\triangleq} \left((\mathsf{com}_b,\mathsf{com}_{1-b}), \mathsf{ch}, (\mathsf{ch}\oplus\mathsf{ch}_{1-b}, \mathsf{resp}_b, \mathsf{resp}_{1-b})\right) \qquad (\because (com,ch,resp) \overset{\mathrm{d}}{\triangleq} (\mathsf{com}_b,\mathsf{ch},\mathsf{resp}_b)) \\
&= \left((\mathsf{com}_b,\mathsf{com}_{1-b}), \mathsf{ch}, (\mathsf{ch}_b,\mathsf{resp}_b,\mathsf{resp}_{1-b})\right) = \left((\mathsf{com}_0,\mathsf{com}_1),\mathsf{ch},(\mathsf{ch}_0,\mathsf{resp}_0,\mathsf{resp}_1)\right) = (\mathsf{com},\mathsf{ch},\mathsf{resp}).
\end{aligned}$$

- **Case 2**: $b = 1$. Likewise, we have

$$
\begin{aligned}
&(\mathsf{com}', \mathsf{ch}', \mathsf{resp}') \\
&= \big((\mathsf{com}'_0, \mathsf{com}'_1), \mathsf{ch}'_0 \oplus \mathsf{ch}'_1, (\mathsf{ch}'_0, \mathsf{resp}'_0, \mathsf{resp}'_1)\big) \\
&= \big((\mathsf{com}'_{1-b}, \mathsf{com}'_b), \mathsf{ch}'_b \oplus \mathsf{ch}'_{1-b}, (\mathsf{ch}'_{1-b}, \mathsf{resp}'_{1-b}, \mathsf{resp}'_b)\big) \\
&\overset{\mathrm{d}}{=} \big((\mathsf{com}_{1-b}, \mathsf{com}'_b), \mathsf{ch}'_b \oplus \mathsf{ch}_{1-b}, (\mathsf{ch}_{1-b}, \mathsf{resp}_{1-b}, \mathsf{resp}'_b)\big) \\
&\qquad (\because (\mathsf{com}_{1-b}, \mathsf{ch}_{1-b}, \mathsf{resp}_{1-b}), (\mathsf{com}'_{1-b}, \mathsf{ch}'_{1-b}, \mathsf{resp}'_{1-b}) \overset{\mathrm{iid}}{\sim} \mathsf{Sim}_{1-b}(x_{1-b})) \\
&\overset{\mathrm{d}}{=} \big((\mathsf{com}_{1-b}, com), ch \oplus \mathsf{ch}_{1-b}, (\mathsf{ch}_{1-b}, \mathsf{resp}_{1-b}, resp)\big) \text{ where } (com, st) \leftarrow P_b(x_b, y), ch \xleftarrow{\$} \{0,1\}^n, resp \leftarrow P_b(st, ch \\
&\qquad (\because (\mathsf{com}'_b, \mathsf{ch}'_b, \mathsf{resp}'_b) \overset{\mathrm{d}}{=} (com, ch, resp) \text{ by } (5.4)) \\
&\overset{\mathrm{d}}{=} \Big((\mathsf{com}_{1-b}, com), ch^\star, (\mathsf{ch}_{1-b}, \mathsf{resp}_{1-b}, resp)\Big) \\
&\qquad\qquad\qquad\qquad \text{where } (com, st) \leftarrow P_b(x_b, y), ch^\star \xleftarrow{\$} \{0,1\}^n, resp \leftarrow P_b(st, ch^\star \oplus \mathsf{ch}_{1-b}) \\
&\quad (\because ch \text{ is fresh, uniform, and independent of } \mathsf{ch}_{1-b} \implies ch^\star \triangleq ch \oplus \mathsf{ch}_{1-b} \text{ is also uniform,} \\
&\qquad \text{so we redefine } ch^\star \xleftarrow{\$} \{0,1\}^n, ch \triangleq ch^\star \oplus \mathsf{ch}_{1-b} \text{ so that } ch^\star \text{ becomes fresh}) \\
&\overset{\mathrm{d}}{=} \big((\mathsf{com}_{1-b}, \mathsf{com}_b), \mathsf{ch}, (\mathsf{ch}_{1-b}, \mathsf{resp}_{1-b}, \mathsf{resp}_b)\big) \qquad (\because (com, ch, resp) \overset{\mathrm{d}}{=} (\mathsf{com}_b, \mathsf{ch}, \mathsf{resp}_b)) \\
&= \big((\mathsf{com}_{1-b}, \mathsf{com}_b), \mathsf{ch}, (\mathsf{ch}_{1-b}, \mathsf{resp}_{1-b}, \mathsf{resp}_b)\big) = ((\mathsf{com}_0, \mathsf{com}_1), \mathsf{ch}, (\mathsf{ch}_0, \mathsf{resp}_0, \mathsf{resp}_1)) = (\mathsf{com}, \mathsf{ch}, \mathsf{resp})\,.
\end{aligned}
$$

In either case, we have $(\mathsf{com}', \mathsf{ch}', \mathsf{resp}') \overset{\mathrm{d}}{=} (\mathsf{com}, \mathsf{ch}, \mathsf{resp})$, as desired.

Therefore, by definition, the (potentially) Sigma protocol (P, V) for $\mathcal{R}_{\mathsf{OR}}$ satisfies HVZK. ⊛

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